

Quasi-Zenith Satellite System
Interface Specification
Signal Authentication Service
(IS-QZSS-SAS-001)

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Cabinet Office

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1 Scope

This document prescribes the interface specifications of Navigation Message Authentication (NMA) of QZSS signals and GNSS signals. Navigation Message Authentication (NMA) is one of the services of QZSS Signal Authentication Services (QZSS-SAS). The interface specification described herein include message specifications, digital signature generation and authentication message broadcast patterns.

2 Relevant Documents and Definition of Terms

2.1 Applicable Documents

2.1.1 GNSS Documents

- [1] IS-QZSS-PNT-005, Quasi-Zenith Satellite System Interface Specification Satellite Positioning, Navigation and Timing Service
- [2] IS-QZSS-MDC-002, Quasi-Zenith Satellite System Interface Specification Multi-GNSS Advanced Orbit and Clock Augmentation - Precise Point Positioning
- [3] IS-GPS-200N, NAVSTAR GPS Space Segment/Navigation User Interfaces
- [4] IS-GPS-705J, NAVSTAR GPS Space Segment/ User Segment L5 Interfaces
- [5] IS-GPS-800J , NAVSTAR GPS Space Segment/ User Segment L1C Interfaces
- [6] Galileo OS SIS 2.0, EUROPEAN GNSS (GALILEO) OPEN SERVICE SIGNAL-IN-SPACE INTERFACE CONTROL DOCUMENT
- [7] PS-QZSS-004, Quasi-Zenith Satellite System Performance Standard (To be published.)

Table 2-1: GNSS IS Document Information

Service Type	Signal Type	Message Type	Document Name	Latest Release Date	Download link
QZSS PNT	L1C/A L1C/B	LNAV	applicable document[1]	January 2021	[1]
	L5	CNAV		January 2021	
	L1C	CNAV2		January 2021	
QZSS MADOCA	L6E	—	applicable document [2]	February 2022	[2]
GPS	L1C/A	LNAV	applicable document [3]	Aug 2022	[3]
	L5	CNAV	applicable document [4]	Aug 2022	[4]
	L1C	CNAV2	applicable document [5]	Aug 2022	[5]
Galileo	E1B	I/NAV	applicable document [6]	January 2021	[6]
	E5a	F/NAV			

2.1.2 Signal Authentication Algorithm Documents

[8] FIPS PUB 186-4, FEDERAL INFORMATION PROCESSING STANDARDS PUBLICATION Digital Signature Standard (DSS)

[9] FIPS PUB 180-4, FEDERAL INFORMATION PROCESSING STANDARDS PUBLICATION Secure Hash Standard (SHS)

Table 2-2: Signal Authentication Algorithm Document Information

Document Name	Purpose	Latest Release Date	Download link
applicable document [8]	Digital Signature Computation	July 2013	[8]
	Polynomial Curve		
applicable document [9]	Hash Function	August 2015	[9]

2.2 Reference Documents

[10] RFC5480, Elliptic Curve Cryptography, Public Key Information

Table 2-3: Reference Document Information

Document Name	Purpose	Latest Release Date	Download link
reference document [10]	DAR file format	March 2009	[10]

2.3 Terms and Definitions

Table 2-4: List of Terms and Definitions

Term	Definition
RNAV	RNAV (Reference Navigation Message) is a portion of navigation message that is used to generate a digital signature
MASK	MASK (Mask Data) is a matrix of the same size as Reference Navigation Message (RNAV) to select necessary data bits from RNAV.
MNAV	MNAV (Mask Navigation Message) is the output data after AND operation between RNAV and MASK
RAND	RAND (Reference Authentication Navigation Data) is the combined data of all MNAV.
HASH	HASH (Hash Message Digest) is output data from Hash function operation on RAND
DS	DS (Digital Signature) is output data from signing process using ECDSA algorithm on HASH.
RDS	RDS (Reformat Digital Signature) is generated by adding Key ID, SALT Data and Reserve data to DS
DSS	RDS is split to DSS (Digital Signature Segment) into smaller segments to fit within the QZSS LNAV, CNAV and CNAV2 messages.
NMAF	NMAF (Navigation Message Authentication Frame) is the navigation message frame that has Digital Signature Segment (DSS) data for message authentication.
RTOW	TOW data in navigation message of GPS and Galileo are modified to RTOW by adding and offset of "X" minutes to generate RNAV for GPS and Galileo
QZNMA	QZNMA (Quasi-Zenith Satellite Navigation Message Authentication) is the message data for the Navigation Message Authentication generated by the QZSS.

2.4 Abbreviations

-A-	AND AUTH	AND Operation Authentication
-B-		
-C-	CNAV CRC	Civil Navigation Cyclic Redundancy Check
-D-	DER DS DSS	Distinguished Encoding Rule Digital Signature Digital Signature Split
-E-	ECC ECDSA ECEF ECI EOP ERD EXOR	Elliptic Curve Cryptography Elliptic Curve Digital Signature Algorithm Earth Center Earth Fixed Earth Centered Inertial Earth Orientation Parameters Estimated Range Deviation Exclusive or
-F-	FEC	Forward Error Correction
-G-	GEO GGTO GPS GPST	Geostationary Orbits time offset between GPST and GNSST Global Positioning System GPS Time
-H-	HASH HOW	Hash Digest Message Hand Over Word
-I-	IERS IODC IODE ISC ISF ITOW ITRF ITRS	International Earth Rotation and Reference Systems Service Issue of Data Clock Issue of Data Ephemeris Inter Signal Correction Integrity Status Flag Interval Time of week International Terrestrial Reference Frame International Terrestrial Reference System
-J-		
-K-		
-L-	LDPC LNAV LSB	Low Density Parity Check Legacy NAVigation Least Significant Bit
-M-	MASK MCS MSB MSG	Mask Data Master Control Station Most Significant Bit Message
-N-	NMA NAV NH Code	Navigation Message Authentication Navigation Message Neuman-Hoffman Code

	NICT	National Institute of Information and Communications Technology
	NMCT	Navigation Message Correction Table
	NTE	Not-To-Exceed
-O-		
	OR	OR Operation
-P-		
	PLL	Phase Locked Loop
	PRN	Pseudorandom Noise
	PUK	Public Key
	PVT	Private Key
-Q-		
	QGEO	Quasi-Geostationary Orbits
	QMS	QZSS Monitoring Station
	QPSK	Quadrature Phase Shift Keying
	QZO	Quasi-Zenith Orbits
	QZS	Quasi-Zenith Satellite
	QZSS	Quasi-Zenith Satellite System
	QZSST	QZSS Time
-R-		
	RAND	Random Authentication Navigation Data
	RDS	Reformat Digital Signature
	RF	Radio Frequency
	RMS	Root Mean Square
	RNAV	Reference Navigation Message
-S-		
	SAS	Signal Authentication Services
	sc/s	semi-circle per second
	SF	Subframe
	SHA	Secure Hashing Algorithm
	SIS	Signal-In-Space
	SIS-URE	Signal-In-Space User Range Error
	sps	symbols per second
	SV	Space Vehicle
-T-		
	TAI	International Atomic Time
	TGD	Timing Group Delay
	TLM Word	Telemetry Word
	TMBOC	Time Multiplex BOC
	TOI	Time of Interval
	TOW	Time of week
	TTA	Time-To-Alert
-U-		
	URA	User Range Accuracy
	USNO	United States Naval Observatory
	UT1	Universal Time1
	UTC	Coordinated Universal Time
-V-		
-W-		
-X-		
	XOR	Exclusive or
-Y-		
-Z-		

3 Signal Authentication Services

QZSS provides signal authentication services for QZSS L1C/A, L1C/B, L1C and L5 signals. It is done by transmitting Navigation Message Authentication (NMA) data embedded into the navigation messages of the respective QZSS signals. A NMA data is a portion of digital signature computed from the navigation message of a signal that has to be authenticated. For example, if QZSS L1C/A signal has to be authenticated, sub-frames 1, 2 and 3 of L1C/A signal are used to compute a digital signature. This digital signature is then reformatted to insert in Sub-frame 5. The satellite broadcasts this signal with NMA data in sub-frame 5. Table 3-1 shows the list of QZSS signals that is used to authenticate the respective QZSS signals. QZSS L1C/A signal is used to authenticate L1C/A signal. Similarly, L1C signal is used to authenticate L1C signal and L5 signal is used to authenticate L5 signal.

QZSS also provides signal authentication services for GPS L1C/A, L1C, L5 signals and Galileo E1B and E5a signals. It is done by transmitting NMA data embedded into the navigation messages of QZSS L6E signal to authenticate GPS and GALILEO signals. Table 3-2 shows the list of GPS and Galileo signals that is authenticated by using QZSS L6E signal.

Table 3-1: QZSS signals that will be used to authenticate respective QZSS navigation messages

SV ID ^{*1,*2}	QZSS Signals with QZSS Navigation Message Authentication				Comments
	PRN ID	Signal	Signal	Signal	
1	193	L1C/A	L1C	L5	Eligible if operational
2	194	L1C/A	L1C	L5	
3	195	L1C/A	L1C	L5	
4	196	L1C/A	L1C	L5	Switching from L1C/A to L1C/B in the future
	203	L1C/B	-	-	
5	197	L1C/A	L1C	L5	
	204	L1C/B	-	-	
7	199	L1C/A	L1C	L5	
8	200	L1C/A	L1C	L5	
	205	L1C/B	-	-	
9	201	L1C/A	L1C	L5	
	206	L1C/B	-	-	

Note

* 1: SV ID referred to IS-QZSS-PNT.

* 2: PRN code assignment including non-standard codes could be updated according to the future development.

Table 3-2: QZSS signal that will be used to authenticate GPS and Galileo navigation messages

SV ID	QZSS L6E Signal with GPS and Galileo Navigation Message Authentication			Comments
	PRN ID	Signal	GNSS Navigation Message	
1	203	-	-	
2	204	L6E	GPS LNAV GPS CNAV GPS CNAV2 Galileo I/NAV Galileo F/NAV	
3	205	L6E		
4	206	L6E		
5	207	L6E		
7	209	L6E		
8	210	L6E		

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9	211	L6E		
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3.1 Overview of Navigation Message Authentication

3.1.1 QZSS Signals

Figure 3-1 shows flowchart of QZSS navigation message authentication process. Message generation center of QZSS selects a portion of QZSS navigation message that will be used for signal authentication. The navigation messages are LNAV, CNAV and CNAV2. Each message is used to generate a RAND message to compute a hash message digest using a hash function. The hash message digest is signed by a private key generated by ECDSA algorithm. The signing process generates a digital signature. The digital signature data is reformatted to fit within the navigation message of a QZSS signal that will be used for authentication. In the case of QZSS L1C/A or C/B, L1C and L5, the digital signatures are broadcasted using navigation messages of QZSS L1C/A or C/B (LNAV), QZSS L1C (CNAV2) and QZSS L5 (CNAV) respectively.

A receiver gets LNAV, CNAV, CNAV2 messages from QZSS satellites. Reference Navigation message (RNAV) and digital signatures (DS) are retrieved from these navigation messages. Reference Authentication Navigation Data (RAND) is generated from RNAV data. Hash message digest is computed from RAND using a Hash function. The hash message digest is used to verify the signal by using the DS data and public key. The result of verification process decides success or failure of navigation message authentication.

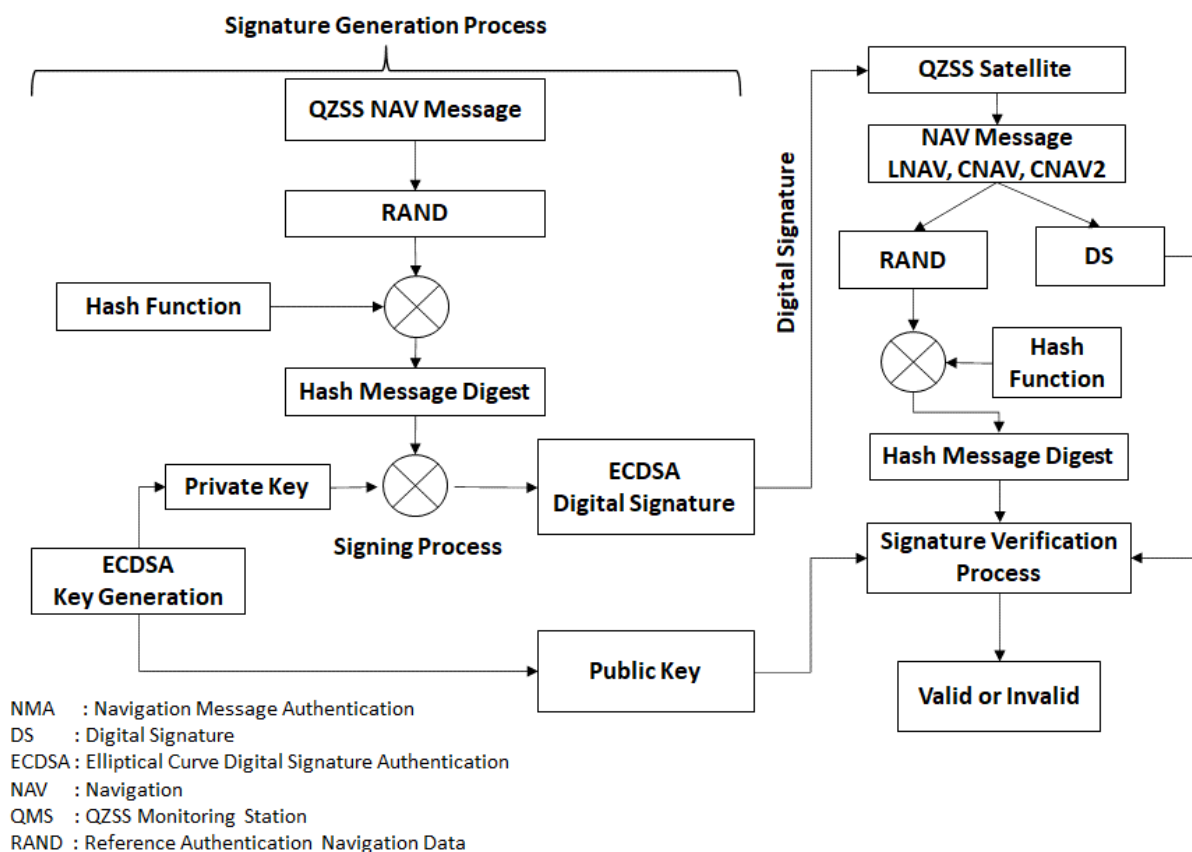


Figure 3-1: Flowchart of QZSS signal authentication

3.1.2 GNSS Signals

QZSS also provides GNSS navigation message authentication for GPS and Galileo navigation signals. Figure 3-2 shows a flowchart of GNSS navigation message authentication using QZSS L6E signal. Authentication is provided for GPS LNAV message of L1C/A signal, CNAV message of L5 signal, CNAV2 message of L1C signal, Galileo I/NAV message of E1B and F/NAV message of E5a.

QZSS provides GNSS navigation message authentication for GPS LNAV, CNAV, CNAV2 messages and Galileo I/NAV, F/NAV messages using QZSS L6E signal. (QZSS provides GNSS navigation message authentication for GPS LNAV, CNAV, CNAV2 messages and Galileo I/NAV, F/NAV messages using QZSS L6E signal. See Table 3-2).QZSS Monitoring Stations (QMS) located in Japan and different places around the world continuously observe GNSS signals. QMS receives navigation messages of GPS and Galileo signals of the same satellites from multiple monitoring stations. The multiple navigation messages received by QMS for the same satellites are evaluated for bit errors and other discrepancies in the navigation messages. If there are no discrepancies in the navigation messages, these data are used as GNSS NAV Message. This GNSS NAV Message is reformatted with additional header information and named as RAND (Reference Authentication Navigation Data). A Hash function is applied on RAND to output Hash Message Digest. The hash message digest is signed using a private key generated by ECDSA algorithm to generate a digital signature. This digital signature data is reformatted and broadcasted using QZSS L6E (MADOCA) signal as QZNMA (QZSS Navigation Message Authentication) message.

A receiver gets LNAV, CNAV, CNAV2 navigation messages from GPS satellites and I/NAV, F/NAV navigation messages from Galileo satellites. The digital signature data that corresponds to GPS and Galileo navigation messages is available from QZSS L6E NMA message. This digital signature with the corresponding public key is used to authenticate GPS and Galileo navigation messages.

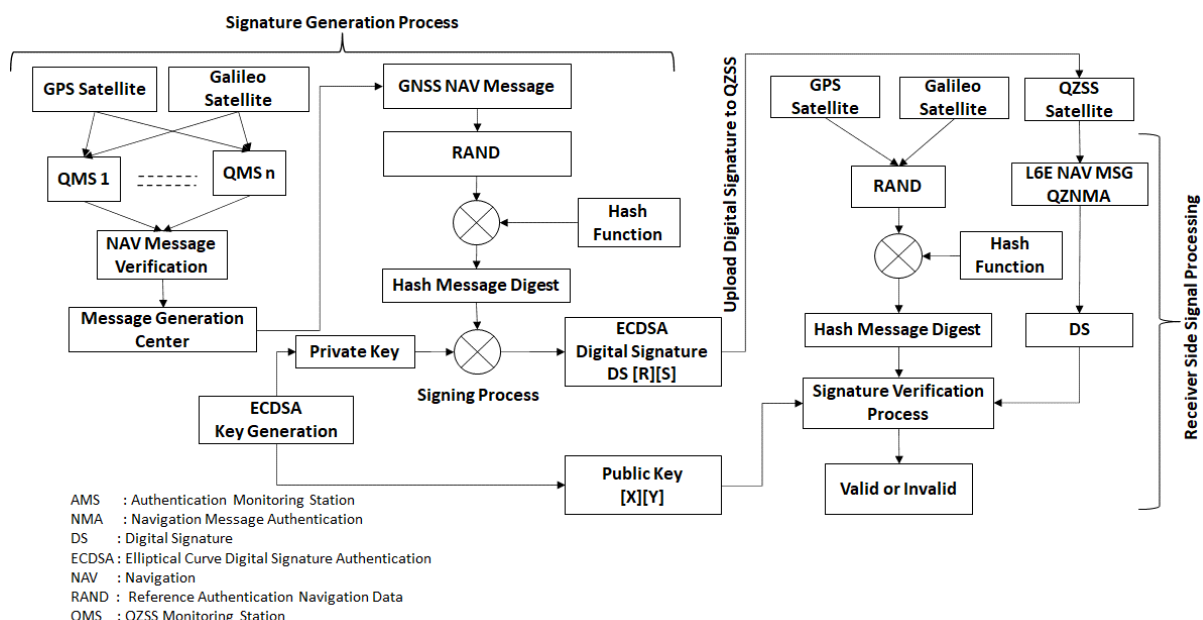


Figure 3-2: Flowchart of GNSS signal authentication using QZSS L6E signal

4 Message Definitions

Definition of terminologies related with message authentication in this document are explained.

4.1 Reference Navigation Message (RNAV)

Reference Navigation Message (RNAV) is a portion of navigation message that is used to generate a digital signature. RNAV is generated from a navigation message of a signal that has to be authenticated. RNAV uses all navigation data bits including parity bits. Table 4-1 shows which navigation messages are used for RNAV. It also shows which navigation messages are used to broadcast digital signature data.

Table 4-1: Reference Navigation Message (RNAV) and Navigation Message Authentication Frame (NMAF)

SV Type	Message Type	Signals	Navigation Message used for RNAV and Data Size	Navigation Message used to broadcast NMAF See 4.11
QZSS	LNAV	L1C/A L1C/B	Sub-frame 1, 300 bits Sub-frame 2, 300 bits Sub-frame 3, 300 bits	QZSS LNAV Sub-Frame 5 SV ID 60
	CNAV	L5	Message Type 10, 300 bits Message Type 11, 300 bits Message Type 30, 300 bits ^{*1}	QZSS CNAV Sub-frame 4 Message Type 60
	CNAV2	L1C	Sub-frame 1, 9 bits Sub-frame 2, 600 bits	QZSS CNAV2 Sub-frame 3 Page ID 60
GPS	LNAV	L1C/A	Sub-frame 1, 300 bits Sub-frame 2, 300 bits Sub-frame 3, 300 bits	QZSS L6E QZNMA Signal Type: 1 Vendor ID: 011
	CNAV	L5	Message Type 10, 300 bits Message Type 11, 300 bits Message Type 32, 300 bits ^{*1}	QZSS L6E QZNMA Signal Type: 2 Vendor ID: 011
	CNAV2	L1C	Sub-frame 1, 9 bits Sub-frame 2, 600 bits	QZSS L6E QZNMA Signal Type: 3 Vendor ID: 011
Galileo	F/NAV	E5a	Page Type 1, 244 bits Page Type 2, 244 bits Page Type 3, 244 bits Page Type 4, 244 bits	QZSS L6E QZNMA Signal Type: 4 Vendor ID: 011
	I/NAV	E1B	Word Type 1, 128 bits Word Type 2, 128 bits Word Type 3, 128 bits Word Type 4, 128 bits Word Type 5, 128 bits	QZSS L6E QZNMA Signal Type: 5 Vendor ID: 011

Note

* 1 : One of the message types 30 or above will be used.

4.2 Mask Data (MASK)

Mask Data (MASK) is a matrix of the same size as Reference Navigation Message (RNAV). It contains "1" for ephemeris and time related bits of RNAV and "0" for parity, CRC and some other data bits. An

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“AND” operation is performed between the RNAV and MASK. The purpose of using MASK is to select only target navigation data bits from RNAV.

4.3 Mask Navigation Message (MNAV)

MNAV is the output data after AND operation between RNAV and MASK. Size of each MNAV is the same as the size of RNAV.

4.4 Reference Authentication Navigation Data (RAND)

Reference Authentication Navigation Data (RAND) is the combined data of all MNAV. In the case of LNAV, all three frames of MNAV are combined into one frame of 900 bits. 8 bits of Key ID, 16 bits of SALT and 4 reserve bits are added to make RAND frame size 928 bits from MNAV.

4.5 Key ID

Key ID refers to a public key ID that is used to verify a digital signature.

4.6 SALT

SALT is a 16-bit random data.

4.7 Hash Message Digest (HASH)

Hash Message Digest (HASH) is output data from Hash function operation on RAND. The size of Hash Message Digest is 256 bits.

4.8 Digital Signature (DS)

Digital Signature (DS) is output data from signing process using ECDSA algorithm on HASH. The size of digital signature is 512 bits.

4.9 Reformat Digital Signature (RDS)

Reformat Digital Signature (RDS) is generated by adding 8 bits of Key ID, 16 bits of SALT Data and 4 bits of Reserve data to Digital Signature (DS) data.

4.10 Digital Signature Segment (DSS)

Since the size of the Reformat Digital Signature (RDS) data is longer than the size of a navigation message format, it is necessary to split RDS into smaller segments to fit within the QZSS LNAV, CNAV and CNAV2 messages. While it is not necessary to split the RDS data for GPS and Galileo since it is broadcasted from QZSS L6E signal. QZSS L6E MADOCA-PPP message has enough space to accommodate RDS data within one L6E message frame.

4.11 Navigation Message Authentication Frame (NMAF)

Navigation Message Authentication Frame (NMAF) is the navigation message frame that has Digital Signature Segment (DSS) data for message authentication. In the case of QZSS LNAV, CNAV and CNAV2, Digital Signature Segment (DSS) data are broadcasted from QZSS LNAV Sub-frame 5, CNAV Sub-frame 4 and CNAV2 Sub-frame 3 respectively. In the case of GPS LNAV, CNAV, CNAV2 and Galileo I/NAV and F/NAV, Digital Signature Segment (DSS) data are broadcasted from QZSS L6E signal.

5 QZSS Navigation Message Authentication

Authentication of QZSS navigation messages LNAV, CNAV and CNAV2 of QZSS L1C/A or L1C/B, L5 and L1C signals are explained.

5.1 QZSS LNAV Message Authentication

5.1.1 Overview

Figure 5-1 shows overview of QZSS LNAV message authentication. Sub-frame 1, 2 and 3 of QZSS LNAV message are used as Reference Navigation Message. All 300 bits of each sub-frame including parity bits are taken. Three Mask Data frames of each 300 bits are used to perform “AND” operation between the Reference Navigation Message and Mask Data. Mask Data frames are used to select time, clock and ephemeris related data from Reference Navigation Message. This operation outputs three frames each with 300 bits. All these frames are combined together to generate a Masked Navigation Message of size 900 bits as shown in Figure 5-8.

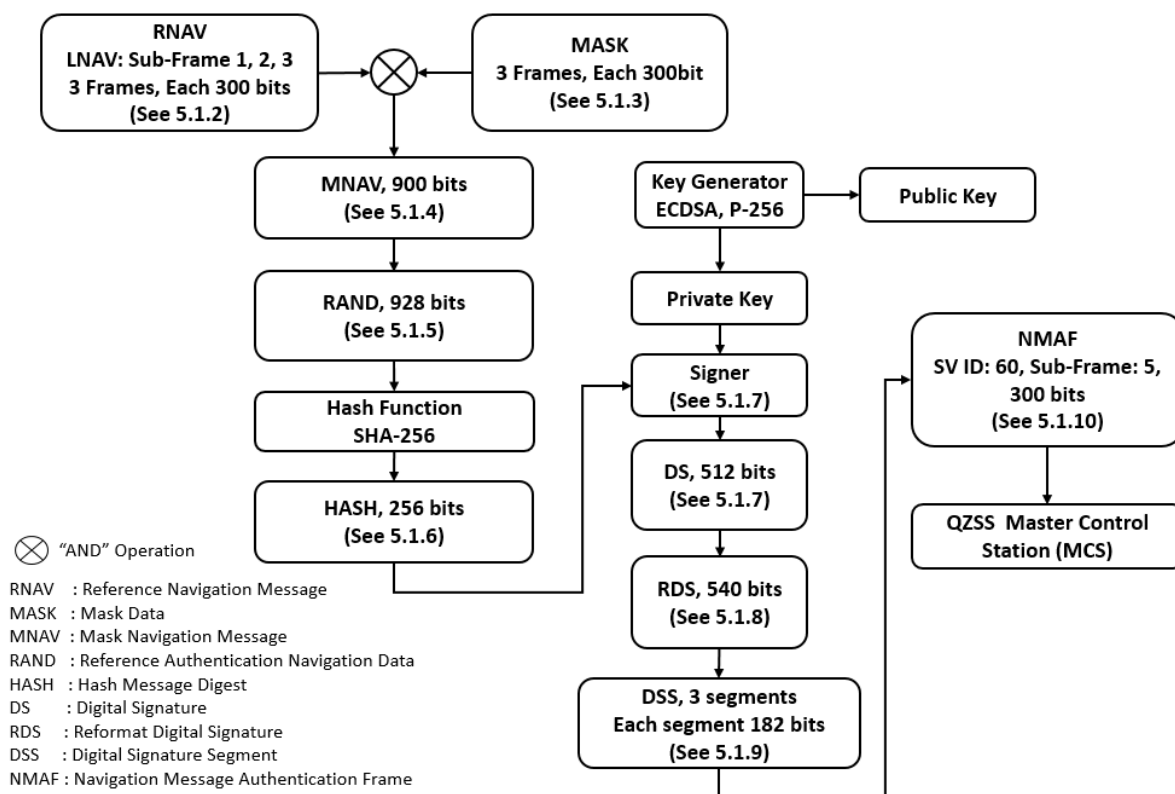


Figure 5-1: QZSS LNAV Signal Authentication Method

Key ID, SALT and Reserve bits are added to Masked Navigation Message to generate Reference Authentication Navigation Data (RAND) of size 928 bits as shown in Figure 5-9. This message is the main input data to compute digital signature. Hash function SHA-256 is used to compute hash message digest of RAND. This generates 256 bits of hash message digest. A private key is used to sign the hash digest message using ECDSA P-256 to generate Digital Signature data. The size of Digital Signature data is 512 bits. This Digital Signature data is reformatted by adding Key ID, SALT and Reserve data to generate 540 bits long Reformat Digital Signature data. Since, 540 bits can't be accommodated into one sub-frame of LNAV, it is necessary to split Reformat Digital Signature data into smaller

segments. This is done by splitting into three segments each of 180 bits. A 2-bit Segment ID is added to each segment to generate three Digital Signature Segment of size 182 bits. Each of these Digital Signature Segment data is embedded into Sub-frame 5 of LNAV. The sub-frame that has authentication related data is called Navigation Message Authentication Frame. This message is uploaded to QZSS satellites.

5.1.2 LNAV Reference Navigation Message (RNAV)

The Reference Navigation Message for QZSS L1C/A and L1C/B signals are taken from LNAV sub-frame 1, 2 and 3. Each sub-frame data is 300 bits long. Figure 5-2, Figure 5-3 and Figure 5-4 show sub-frame 1, 2 and 3 data of QZSS L1C/A, L1C/B used for Reference Navigation Message. Refer applicable document [1] for details on QZSS LNAV navigation message details.

Note that only the non-shaded portions of the sub-frame are used to generate digital signature. The shaded portions of the navigation message are not used. This applies to all other navigation messages as well.

MSB																														LSB
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
WORD1	PREAMBLE								TLM Message														ISF		R	Parity				
WORD2	TOW-Count Message																	Alert		AS	SubFrameID		P.C.Set		Parity					
WORD3	WN									Code on L2			URA INDEX				SV HEALTH				IODC		Parity							
WORD4	TOW-Count Flag	Reserved																								Parity				
Reserved																								Parity						
Reserved																								Parity						
Reserved																								Parity						
WORD7	Reserved															T _{GD}					Parity									
WORD8	IODC								T _{oc}																	Parity				
WORD9	a _{f2}								a _{f1}																	Parity				
WORD10	a _{f0}																	P.C.Set		Parity										

Figure 5-2: QZSS LNAV Sub-frame 1 navigation message

MSB																														LSB				
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30				
WORD1	PREAMBLE								TLM Message														ISF		R	Parity								
WORD2	TOW-Count Message																	Alert		AS	SubFrameID		P.C.Set		Parity									
WORD3	IODE								C _{rs}														Parity											
WORD4	Δn								M ₀ (24 LSBs)										M ₀ (8 MSBs)										Parity					
WORD5	M ₀ (24 LSBs)																	M ₀ (8 MSBs)													Parity			
WORD6	e(24 LSBs)																	e(8 MSBs)													Parity			
WORD7	e(24 LSBs)																	e(8 MSBs)													Parity			
WORD8	√A(24 LSBs)																	√A(8 MSBs)													Parity			
WORD9	√A(24 LSBs)																	√A(8 MSBs)													Parity			
WORD10	t _{oc}																	FIF		AODO		P.C.Set		Parity										

Figure 5-3: QZSS LNAV Sub-frame 2 navigation message

MSB																												LSB				
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30		
WORD1	PREAMBLE								TLM Message														ISF		R	Parity						
WORD2	TOW-Count Message																	Alert	AS	SubFrameID(2)		P.C.Set		Parity								
WORD3	C _{ic}														Ω ₀ (8 MSBs)										Parity							
WORD4	Ω ₀ (24 LSBs)																											Parity				
WORD5	C _{is}														i ₀ (8 MSBs)										Parity							
WORD6	i ₀ (24 LSBs)																											Parity				
WORD7	C _{rc}														ω(8 MSBs)										Parity							
WORD8	ω(24 LSBs)																											Parity				
WORD9	Ω																												Parity			
WORD10	IODE								IDOT														P.C.Set		Parity							

Figure 5-4: QZSS LNAV Sub-frame 3 navigation message

5.1.3 LNAV MASK Data (MASK)

Figure 5-5, Figure 5-6 and Figure 5-7 show Mask Data for sub-frame 1, 2 and 3 of QZSS LNAV message respectively. Mask Data is used to perform “AND” operation with RNAV to select only required data bits from QZSS LNAV navigation message. The Mask Data has “1” where RNAV data bits are used. The Mask Data has “0” where RNAV data bits are not used.

MSB																														LSB
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
WORD1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
WORD2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WORD3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD10	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0

Figure 5-5: MASK for QZSS LNAV Sub-frame 1

MSB																														LSB
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
WORD1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
WORD2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
WORD3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD10	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0

Figure 5-6: MASK for QZSS LNAV Sub-frame 2

MSB																														LSB
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
WORD1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
WORD2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WORD3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD10	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

Figure 5-7: MASK for QZSS LNAV Sub-frame 3

5.1.4 LNAV Mask Navigation Message (MNAV)

Mask Navigation Message is generated by “AND” operation between Reference Navigation Message and Navigation Mask Data. This generates three 300-bit frames for QZSS LNAV message. The total size of these three frames is 900bits as shown in Figure 5-8.

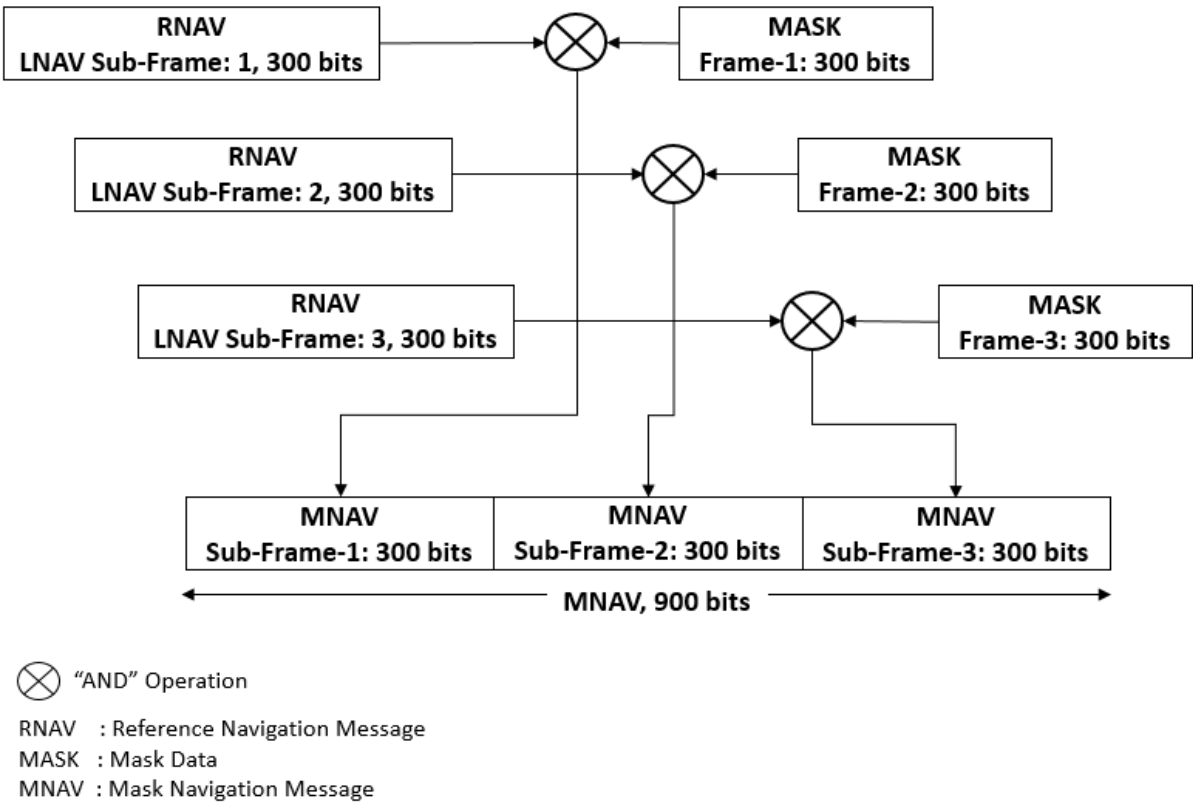


Figure 5-8: Generation of QZSS LNAV MNAV

5.1.5 LNAV Reference Authentication Navigation Data (RAND)

Reference Authentication Navigation Data is generated from Masked Navigation Message generated as shown in Figure 5-8 . Key ID, SALT data and Reserve data are added to Masked Navigation Message to generate Reference Authentication Navigation Data as shown in Figure 5-9.

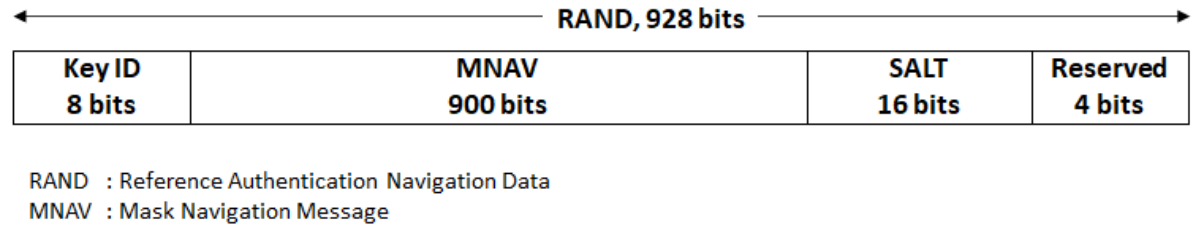


Figure 5-9: Format of QZSS LNAV RAND

Table 5-1: Description of QZSS LNAV RAND Message

Parameter	Description	Length [bits]	Effective Range		Remarks
			min	max	
Key ID	ID of a public key used to verify a digital signature defined in section 4.5	8	1	108	MSB is fixed to zero Only the 7 LSBs are used
MNAV	Output of AND operation between RNAV and MASK. See 4.3 and Figure 5-8	900	—	—	
SALT	A random value generated from true random numbers. See 4.6	16	—	—	A different value is set each time a digital signature is generated.
Reserved	Reserved Bit	4	—	—	All bits are set to zero.

5.1.6 Hash Message Digest (HASH)

Hash message digest is computed using a hash function based on SHA-256.

See 7.2 Hash Method for details on Hash function and Hash message digest.

5.1.7 Digital Signature (DS)

Digital signature is generated by using ECDSA algorithm. See 7.1 Digital Signature and 7.1.2 Polynomial Curve for details on algorithms related with digital signature generation.

5.1.8 LNAV Reformat Digital Signature (RDS)

The 512-bit digital signature data is reformatted as shown in Figure 5-10. 8-bit Key ID, 16-bit SALT and 4 reserve bits are added to 512-bit digital signature data. The total size of reformatted digital signature data is 540 bits.

Table 5-1 shows QZSS LNAV RDS message format.



RDS : Reformat Digital Signature
DS : Digital Signature

Figure 5-10: Format of QZSS LNAV RDS Message

Table 5-2: Descriptions of QZSS LNAV RDS parameters

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
Key ID	ID of a public key used to verify a digital signature defined in section 4.5	8	1	108	MSB is fixed to zero Only the 7 LSBs are used Same as in Table 5-1
DS	Digital signature data generated with the private key corresponding to the KEY ID based on the reference data	512	—	—	Same as in Table 5-1
SALT	A random value generated by a true random number generator, defined in section 4.6	16	—	—	Same as in Table 5-1
Reserved	Reserved Bit	4	—	—	All bits are set to zero.

5.1.9 LNAV Digital Signature Segment (DSS)

The 540 bits of reformatted digital signature message is split into three segments of each 180 bits. A 2-bit Segment ID is added to each segment which makes each segment data 182 bits as shown in Figure 5-11 and described in Table 5-3.

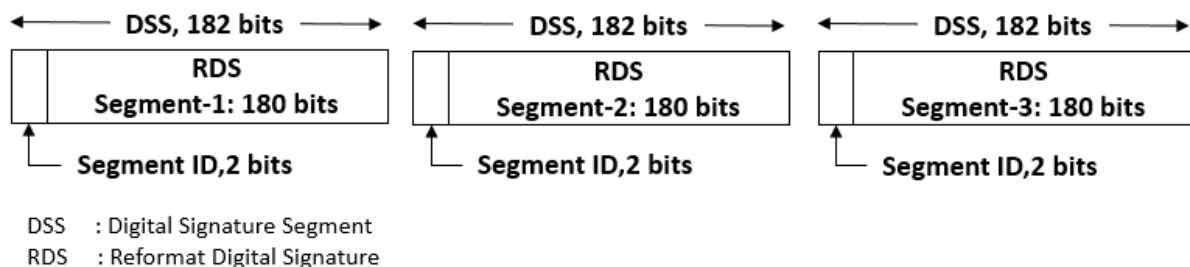


Figure 5-11: QZSS LNAV DSS Data Format

Table 5-3: Description of QZSS LNAV DSS Parameters

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
Segment ID	ID to indicate the corresponding segment data	2	1	3	00b: Unused 01b: Segment 1 10b: segment 2 11b: Segment 3
RDS	Digital Signature data split into three segments	180	—	—	

5.1.10 LNAV Navigation Message Authentication Frame (NMAF)

The Digital Signature Segment (DSS) data shown in Figure 5-11 are inserted into the navigation message of QZSS LNAV sub-frame 5 into each word from Word 3 to Word 10 as shown in Figure 5-12.

Data ID and SV ID indicate whether the sub-frame has a digital signature segment data in the LNAV message. Table 5-4 shows DATA ID and SV ID values when a sub-frame 5 LNAV message has digital signature data.

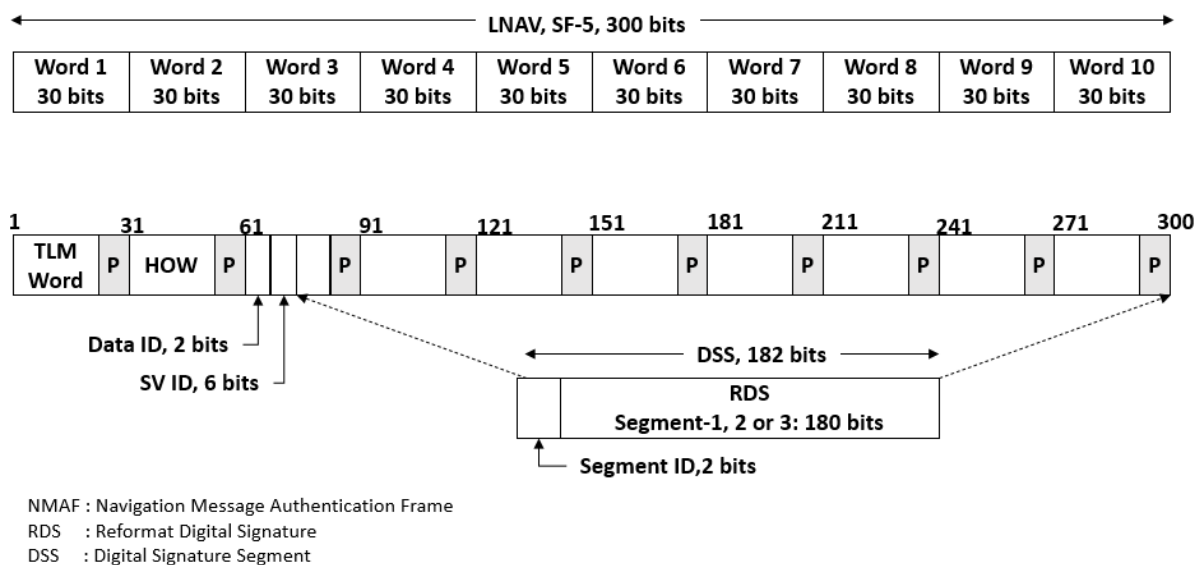


Figure 5-12: QZSS LNAV NMAF message format

Table 5-4: DATA ID and SV ID description

Parameter	Description	Length [bits]	Effective Range		Default Value	Remarks
			Min	Max		
DATA ID	DATA ID in sub-frame 5	2	1	3	3	DATA ID = 3 may be present on other NAV MSG frames.
SV ID	SV ID in sub-frame 5	6	—	—	60	
DSS	Digital Signature data split into three segments	182	—	—	—	

5.1.11 LNAV NMAF Broadcast Pattern

Table 5-5 shows broadcast pattern of one Digital Signature (DS) data that consists of three DSS data formatted as NMAF (see Figure 5-12). These three NMAF broadcasts occur every hour from $N \times 240$ seconds to $180 + N \times 240$ seconds ($N = 0$ to 14). In general, RNAV is SF1, SF2, and SF3, which are subsequently broadcast for 18 seconds from $180 + N \times 240$. Each broadcast corresponds to authentication data for one signal type (L1C/A or L1C/B) of one of the QZSS satellites. 15 broadcasts have 45 Digital Signature Segments. In order to authenticate one of the LNAV signals, it is necessary to receive all three digital signature segments. It takes at least 180 seconds or 3 minutes to get digital signature data for one of the LNAV messages of QZSS L1C/A or L1C/B signal. However, it takes 240 seconds or 4 minutes to get the next digital signature data. This additional time of 60 seconds is due to broadcasting of Ionospheric (Japan area) and UTC parameters with SV ID = 61 after the completion of broadcasting of each digital signature of LNAV message.

Table 5-5 : LNAV NMAF Broadcast Pattern

time [sec]	Subframe ID	Subframe ID	Subframe ID	Subframe ID	Subframe ID
$N \times 240$	Subframe 1	Subframe 2	Subframe 3	Subframe 4	Subframe 5
$30 + N \times 240$	Subframe 1	Subframe 2	Subframe 3	Subframe 4	Subframe 5 SV ID=60, NMAF (1/3)
$60 + N \times 240$	Subframe 1	Subframe 2	Subframe 3	Subframe 4	Subframe 5
$90 + N \times 240$	Subframe 1	Subframe 2	Subframe 3	Subframe 4	Subframe 5 SV ID=60, NMAF (2/3)
$120 + N \times 240$	Subframe 1	Subframe 2	Subframe 3	Subframe 4	Subframe 5
$150 + N \times 240$	Subframe 1	Subframe 2	Subframe 3	Subframe 4	Subframe 5 SV ID=60, NMAF (3/3)
$180 + N \times 240$	Subframe 1	Subframe 2	Subframe 3	Subframe 4	Subframe 5

$N = 0$ to 14

* : Shading represents Navigation Message used for RNAV

5.2 QZSS CNAV Message Authentication

5.2.1 Overview

Figure 5-13 shows overview of QZSS CNAV message authentication. Message Type 10, 11 and one of the message types 30 or above are used as Reference Navigation Message. All 300 bits of each frame including parity bits are taken. Three Mask Data frames of each 300bit size are used to perform “AND” operation between the Reference Navigation Message and Mask Data. Mask Data frames are used to select time, clock and ephemeris related data from Reference Navigation Message. This outputs three frames each with 300 bits.

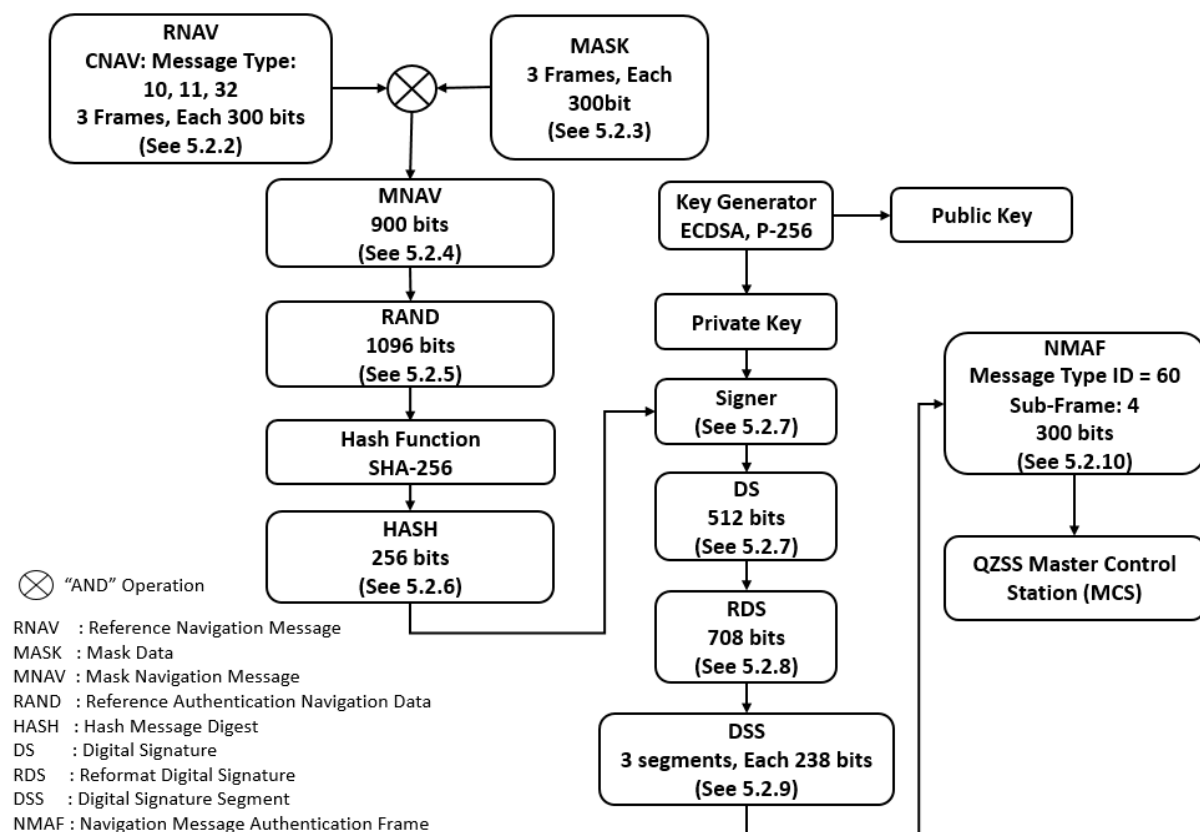


Figure 5-13: Overview of CNAV Message Authentication

These frames are combined together to generate a Masked Navigation Message (MNAV) of size 900 bits as shown in Figure 5-20. Key ID, SALT and Reserve bits are added to Mask Navigation Message (MNAV) to generate Reference Authentication Navigation Data (RAND) of size 1096 bits as shown in Figure 5-21. This message is the main input data to compute digital signature. Hash function SHA-256 is used to compute Hash message digest of RAND. This generates 256 bits of Hash message digest. A private key is used to sign the Hash message digest using ECDSA P-256 algorithm to generate Digital Signature data. The size of Digital Signature data is 512 bits. This Digital Signature data is reformatted by adding Key ID, SALT and Reserve data to generate 708 bits long Reformat Digital Signature (RDS) data. Since, 708 bits can't be accommodated into one sub-frame of CNAV, it is necessary to split Reformat Digital Signature data into smaller sections. This is done by splitting into three segments each of 236 bits. A 2-bit Segment ID is added to each segment to generate three Digital Signature

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Segment of size 238 bits. Each of these Digital Signature Segment data is embedded into Sub-frame 4 of CNAV. The sub-frame that has authentication related data is called Navigation Message Authentication Frame (NMAF). This message is uploaded to QZSS satellites.

5.2.2 CNAV Reference Navigation Message (RNAV)

The Reference Navigation Message for QZSS L5 signal is taken from CNAV Message sub-frame 1, 2 and 3. Each frame is 300 bits long. Figure 5-14: , Figure 5-15: and Figure 5-16: show sub-frame 1, 2 and 3.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30										
1	PREAMBLE(8bit)								PRN(6bit)						Message Type ID(6bit)						Message TOW Count(17bit)																			
2	Message TOW Count(17bit)								A(1bit)	WNn(13bit)												L2 (16bit)			L2 (16bit)			L2 (16bit)			top(11bit)									
3	top					URAd index(5bit)					toe(11bit)										Δ A(26bit)																			
4	Δ A(26bit)														dot A(25bit)																									
5	dot A(25bit)														Δ n0(17bit)																									
6	Δ dot n0(23bit)														M0-n(33bit)																									
7	M0-n(33bit)														en(33bit)																									
8	en(33bit)														en(33bit)																									
9	ωn(33bit)														ωn(33bit)																									
10	ωC(33bit)		L2 (16bit)		L2 (16bit)		Reserved(3bit)			CRC(24bit)																														

Figure 5-14: QZSS CNAV Sub-frame 1

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30		
1	PREAMBLE(8bit)								PRN(6bit)						Message Type ID(6bit)						Message TOW Count(17bit)											
2	Message TOW Count(17bit)								Align 13bit	toe(11bit)											Ωo-n(33bit)											
3	Ωo-n(33bit)														io-n(33bit)																	
4	io-n(33bit)																									Δ dot Ω(17bit)						
5	Δ dot Ω(17bit)											Cis-n(16bit)														Cis-n(16bit)						
6	Cis-n(16bit)											Cic-n(16bit)														Cic-n(16bit)						
7												Crs-n(24bit)											Crc-n(24bit)									
8	Crc-n(24bit)																									Cus-n(21bit)						
9	Cus-n(21bit)								Cuc-n(21bit)														Cus-n(21bit)									
10	Reserved(7bit)						CRC(24bit)																									

Figure 5-15: QZSS CNAV Sub-frame 2

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30			
1	PREAMBLE(8bit)								PRN(6bit)						Message Type ID(6bit)						Message TOW Count(17bit)												
2	Message TOW Count(17bit)								af0-n(10bit)	top(11bit)										URANED0 INDEX(5bit)						URANED1 INDEX(3bit)				URANED2 INDEX(3bit)			
3	toc(11bit)										af0-n(26bit)																						
4	af0-n								af1-n(20bit)																			af2-n					
5	af2-n(10bit)								tEOP(16bit)														PM-X										
6	PM-X(21bit)														PM-dotX(15bit)														PM-Y				
7	PM-Y(21bit)														PM-dotY(15bit)																		
8	PM-dotY						$\Delta UT1(31bit)$																										
9	$\Delta UT1$						$\Delta UdotT1(19bit)$																		Reserved(11bit)								
10	Reserved						CRC(24bit)																										

Figure 5-16: QZSS CNAV Sub-frame 3

5.2.3 CNAV Mask Data (MASK)

Figure 5-17: , Figure 5-18: and Figure 5-19: show Mask Data for Sub-frame 1, 2 and 3 of QZSS CNAV message respectively. Mask Data is used to perform “AND” operation with RNAV to select only required data bits from QZSS CNAV navigation message. The Mask Data has “1” where RNAV data bits are used. The Mask Data has “0” where RNAV data bits are not used.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
10	1	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 5-17: MASK for QZSS CNAV Sub-frame 1

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
10	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 5-18: MASK for QZSS CNAV Sub-frame 2

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 5-19: MASK for QZSS CNAV Sub-frame 3

5.2.4 CNAV Mask Navigation Message (MNAV)

Mask Navigation Message is generated by “AND” operation between Reference Navigation Message and Navigation Mask Data. This generates three 300-bit frames for QZSS CNAV message. The total size of these three frames is 900bits as shown in Figure 5-20.

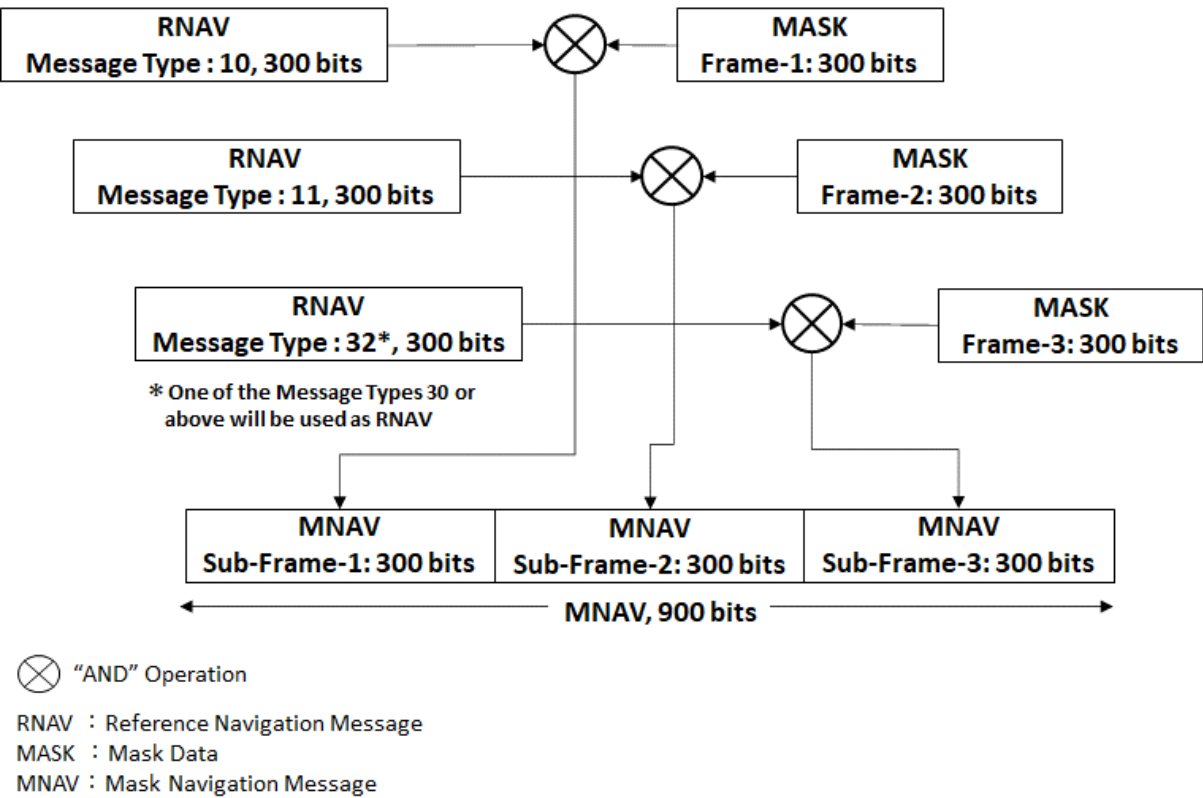


Figure 5-20: Generation of QZSS CNAV MNAV

5.2.5 CNAV Reference Authentication Navigation Data (RAND)

Reference Authentication Navigation Data is generated from Masked Navigation Message generated as shown in Figure 5-20. Key ID, SALT data and Reserve data are added to Masked Navigation Message to generate Reference Authentication Navigation Data as shown in Figure 5-21.

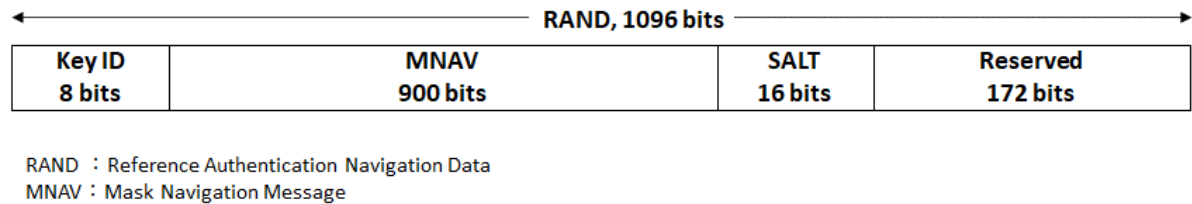


Figure 5-21: Format of QZSS CNAV RAND

Table 5-6: Parameter Description of QZSS CNAV RAND

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
Key ID	ID of a public key used to verify a digital signature. See 4.5	8	1	108	MSB is fixed to zero Only the 7 LSBs are used
MNAV	Output of AND operation between Reference Navigation Message and Mask Data as shown in Figure 5-20:	900	—	—	
SALT	A random value generated from true random numbers.	16	—	—	A different value is set each time a digital signature is generated.
Reserved	Reserved Bit	172	—	—	All bits are set to zero.

5.2.6 Hash Message Digest (HASH)

Hash message digest is computed using a hash function based on SHA-256.

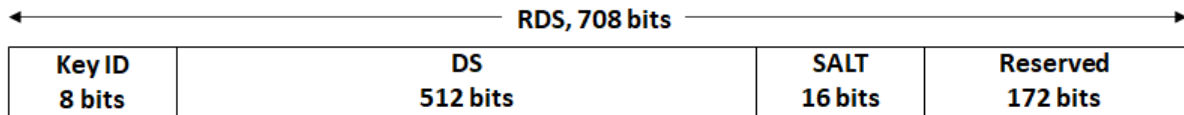
See 7.2 Hash Method for details on Hash function and Hash message digest.

5.2.7 Digital Signature (DS)

Digital signature is generated by using ECDSA algorithm. See 7.1 Digital Signature and 7.1.2 Polynomial Curve for details on algorithms related with digital signature generation.

5.2.8 CNAV Reformat Digital Signature (RDS)

The 512-bit digital signature data is reformatted as shown in Figure 5-22. 8-bit Key ID, 16-bit SALT and 172 reserve bits are added to 512-bit digital signature data. The total size of reformatted digital signature data is 708 bits. Table 5-7 describes QZSS CNAV RDS parameters.



RDS : Reformat Digital Signature
DS : Digital Signature

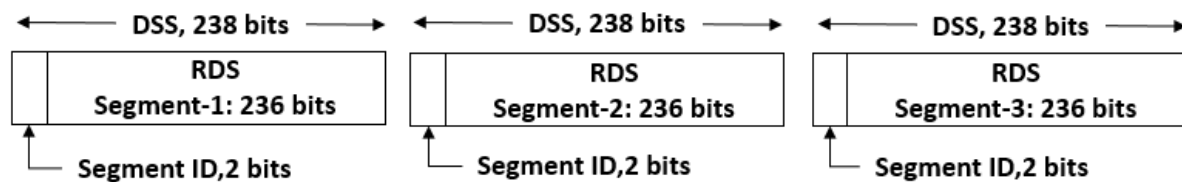
Figure 5-22: Format of QZSS CNAV RDS

Table 5-7: Parameter Description of QZSS CNAV RDS

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
Key ID	ID of a public key used to verify a digital signature. See 4.5	8	1	108	MSB is fixed to zero Only the 7 LSBs are used Same as in Table 5-6
DS	Digital signature data generated with the private key corresponding to the KEY ID based on the reference data.	512	—	—	
SALT	A random value is generated from true random numbers.	16	—	—	A different value is set each time a digital signature is generated.
Reserved	Reserved Bit	172	—	—	All bits are set to zero.

5.2.9 CNAV Digital Signature Segment (DSS)

The 708 bits of reformatted digital signature message is split into three segments of each 236 bits. A 2-bit Segment ID is added to each segment which makes each segment 238 bits as shown in Figure 5-23 and described in Table 5-8.



DSS : Digital Signature Segment
RDS : Reformat Digital Signature

Figure 5-23 QZSS CNAV DSS Format

Table 5-8: QZSS CNAV DSS Parameter Description

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
Segment ID	ID to indicate the corresponding segment data	2	1	3	00b: Unused 01b: Segment 1 10b: segment 2 11b: Segment 3
RDS	Digital Signature data split into three segments	236	—	—	

5.2.10 CNAV Navigation Message Authentication Frame (NMAF)

The Digital Signature Segment data shown in Figure 5-23 are inserted into the navigation message of QZSS CNAV sub-frame 4 as shown in Figure 5-24.

Message Type ID indicates whether the sub-frame has a digital signature segment data in the CNAV message. Table 5-9 shows Message Type ID value when a CNAV Message has a Digital Signature data.

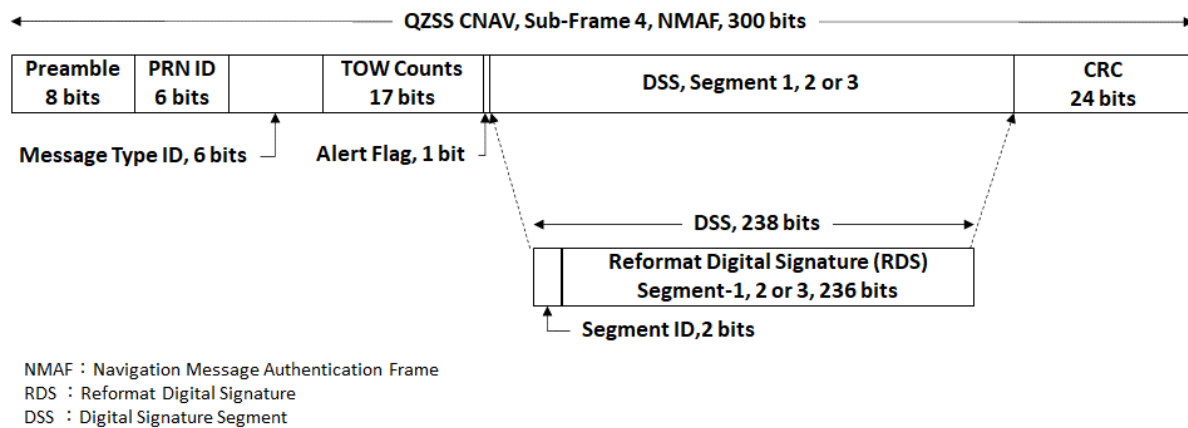


Figure 5-24: Format of QZSS CNAV NMAF

Table 5-9: Description of QZSS CNAV NMAF Parameters

Parameter	Description	Length [bits]	Effective Range		Default Value	Remarks
			Min	Max		
Preamble	Preamble used in CNAV	8	—	—	—	
PRN ID	PRN ID used in CNAV message	6	—	—	—	
Message Type ID	Message Type used in CNAV	6	—	—	60	Message Type ID 60 indicates frame with digital signature data for authentication
TOW Counts	TOW used in CNAV	17	—	—	—	
DSS	DSS data split into three segments	238	—	—	—	
CRC		24	—	—	—	

5.2.11 CNAV NMAF Broadcast Pattern

Table 5-10 shows QZSS L5 navigation message broadcast pattern for sub-frames 1 to 4. Sub-frame 4 broadcasts Digital Signature Segment data (DSS) by replacing Ionosphere and UTC parameters in CNAV as shown in Table 5-10.

Table 5-10 shows the broadcast pattern of one Digital Signature data (QZSS CNAV NMAF 1/3 to 3/3). The broadcast of one digital signature (3 NMAF) takes 168 seconds and occurs 15 times (45 NMAF) every hour.

To authenticate one of the CNAV signals, it is necessary to receive all three digital signature segments. It takes at least 168 seconds to get the digital signature data of one CNAV message of the QZSS L5 signal. However, it can take up to 288 seconds to retrieve the next digital signature data. This additional time of 144 seconds is because broadcast of other message types take place after the broadcast of each digital signature of the CNAV message.

Table 5-10: QZSS CNAV NMAF Broadcast Pattern

Time [sec]	Message Type	Message Type	Message Type	Message Type
T	Other Message Type	Other Message Type	Other Message Type	Message Type 60 NMAF (1/3)
T+24	Other Message Type	Other Message Type	Other Message Type	Other Message Type
T+48	Other Message Type	Other Message Type	Other Message Type	Other Message Type
T+72	Other Message Type	Other Message Type	Other Message Type	Message Type 60 NMAF (2/3)
T+96	Other Message Type	Other Message Type	Other Message Type	Other Message Type
T+120	Other Message Type	Other Message Type	Other Message Type	Other Message Type
T+144	Other Message Type	Other Message Type	Other Message Type	Message Type 60 NMAF (3/3)
T+168	Message Type 10	Message Type 11	Message Type 32	Other Message Type
T+192	Other Message Type	Other Message Type	Other Message Type	Other Message Type

T= 0, 216, 432, 720, 936, 1152, 1440, 1656, 1872, 2160, 2376, 2592, 2880, 3096, 3312 (every hour)

* : Shading represents Navigation Message used for RNAV

5.3 QZSS CNAV2 Message Authentication

5.3.1 Overview

Figure 5-25 shows overview of QZSS CNAV2 message authentication. CNAV2 sub-frame 1 and 2 are used as Reference Navigation Message. 9-bits of sub-frame 1 and 600-bits of sub-frame 2 including parity bits are taken. Two Mask Data frames of 9-bits and 600-bits each 300bit are used to perform “AND” operation between the Reference Navigation Message and Mask Data. Mask Data frames are used to select time, clock and ephemeris related data from Reference Navigation Message. This outputs two frames of 9-bits and 600-bits.

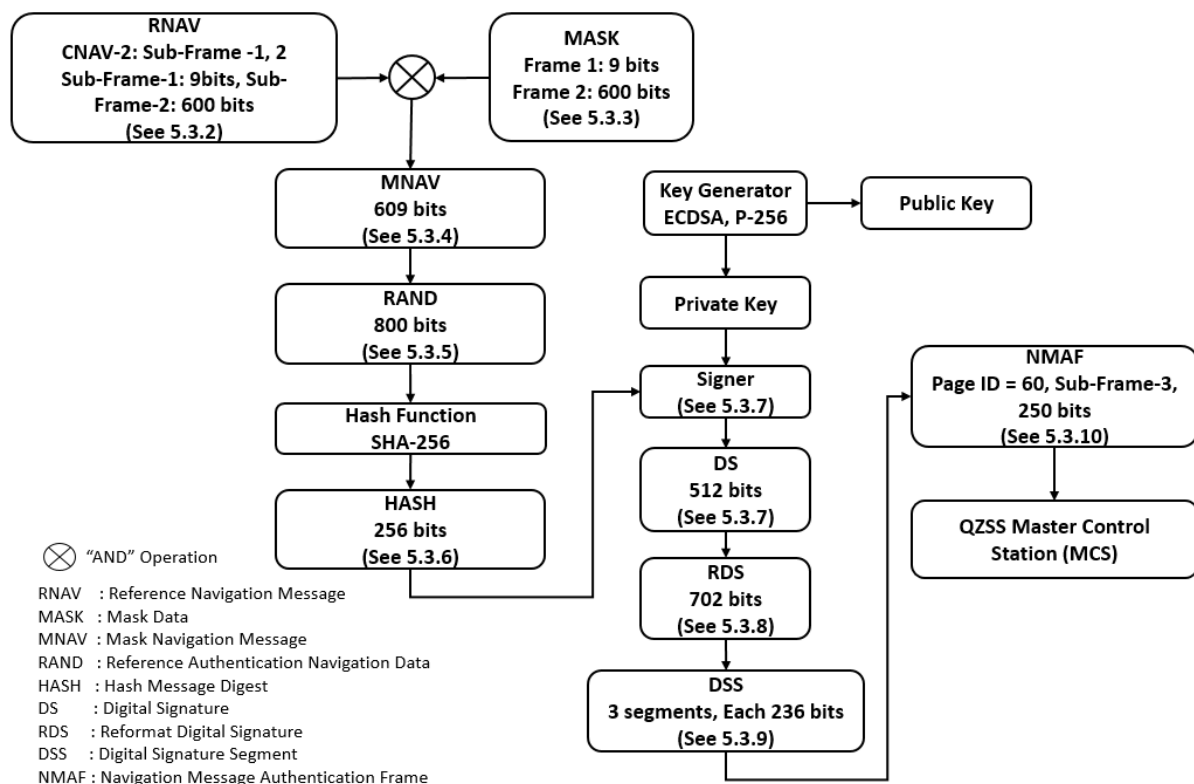


Figure 5-25: Overview of CNAV2 Message Authentication

These frames are combined together to generate a Masked Navigation Message (MNAV) of size 900 bits as shown in Figure 5-30. Key ID, SALT and Reserve bits are added to Mask Navigation Message (MNAV) to generate Reference Authentication Navigation Data (RAND) of size 800 bits as shown in Figure 5-31. This message is the main input data to compute digital signature. Hash function SHA-256 is used to compute Hash message digest of RAND. This generates 256 bits of Hash message digest. A private key is used to sign the Hash message digest using ECDSA P-256 algorithm to generate Digital Signature data. The size of Digital Signature data is 512 bits. This Digital Signature data is reformatted by adding Key ID, SALT and Reserve data to generate 702 bits Reformat Digital Signature (RDS) data. Since, 702 bits can't be accommodated into one sub-frame of CNAV2, it is necessary to split Reformat Digital Signature (RDS) data into smaller sections. This is done by splitting into three segments each of 234 bits. A 2-bit Segment ID is added to each segment to generate three Digital Signature Segment (DSS) of size 236 bits. Each of these Digital Signature Segment (DSS) data is embedded into Sub-frame

3 of CNAV2. The sub-frame that has authentication related data is called Navigation Message Authentication Frame (NMAF). This message is uploaded to QZSS satellites.

5.3.2 CNAV2 Reference Navigation Message (RNAV)

The Reference Navigation Message (RNAV) for QZSS L1C signal is taken from CNAV2 Message sub-frame 1 and 2. Sub-frame 1 has 0 bits and sub-frame 2 has 600 bits as shown in Figure 5-26 and Figure 5-27.

	1	2	3	4	5	6	7	8	9
1	TOI(9bit)								

Figure 5-26: QZSS CNAV2 Sub-frame 1 navigation message

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
1	WN(13bit)													ITOW(8bit)								top(11bit)									
2	top(11bit)		LC length		URA ED INDEX(5bit)					toe(11bit)											ΔA(26bit)										
3	ΔA(26bit)													dot A(25bit)																	
4	dot A(25bit)										Δ n0										Δ dot n0(23bit)										
5	Δ dot n0(23bit)															M 0-n(33bit)										en(33bit)					
6	M 0-n(33bit)																							en(33bit)							
7	en(33bit)																											ωn(33bit)			
8	ωn(33bit)																													Ω 0-n	
9	Ω 0-n(33bit)																														
10	Ω 0-n(33bit)		i 0-n(33bit)																												
11	i 0-n(33bit)					Δ dot Ω(17bit)																	i 0- dot n(15bit)								
12	i 0- dot n(15bit)							C is-n(16bit)													C ic-n(16bit)										
13	C ic-n(16bit)									C rs-n(24bit)																					
14	C rs-n(24bit)				C rc-n(24bit)																										C us-n(21bit)
15	C us-n(21bit)															C uc-n(21bit)															
16	C uc-n(21bit)								URA NED0 INDEX(5bit)					URA NED1 INDEX(3bit)			URA NED2 INDEX(3bit)			a f0-n(26bit)											
17	a f0-n(26bit)															A f1-n(20bit)															
18	a f0-n(26bit)							a f2-n(10bit)										TGD(13bit)													WN OP(8bit)
19	ISC L1CP(13bit)												ISC L1CD(13bit)																		WN OP(8bit)
20	WN OP(8bit)			Reserved(2bit)		CRC(24bit)																									

Figure 5-27: QZSS CNAV2 Sub-frame 2 navigation message

5.3.3 CNAV2 Mask Data (MASK)

Figure 5-28 and Figure 5-29 show Mask Data for sub-frame 1 and 2 of QZSS CNAV2 message respectively. Mask Data is used to perform “AND” operation with RNAV to select only required data bits from QZSS CNAV2 message. The Mask Data has “1” where RNAV data bits are used. The Mask Data has “0” where RNAV data bits are not used.

	1	2	3	4	5	6	7	8	9
1	1	1	1	1	1	1	1	1	1

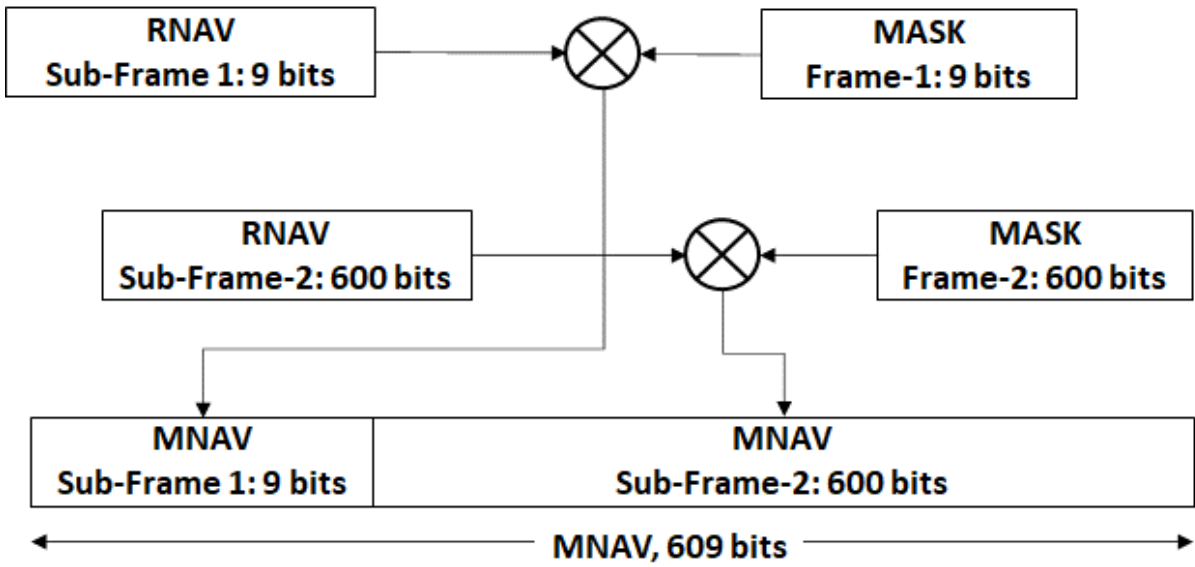
Figure 5-28: MASK for QZSS CNAV2 Sub-frame 1

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
10	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
11	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
12	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
13	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
14	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
15	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
16	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
17	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
18	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
19	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1
20	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 5-29: MASK for QZSS CNAV2 Sub-frame 2

5.3.4 CNAV2 Mask Navigation Message (MNAV)

Mask Navigation Message (MANV) is generated by “AND” operation between Reference Navigation Message (RNAV) and Navigation Mask (MASK) Data. This generates 9-bits frame and 600-bits frame for QZSS CNAV2 message. The total size of these two frames is 609 bits as shown in Figure 5-30.



⊗ “AND” Operation

RNAV : Reference Navigation Message

MASK : Mask Data

MNAV : Mask Navigation Message

Figure 5-30: Generation of QZSS CNAV2 MNAV

5.3.5 CNAV2 Reference Authentication Navigation Data (RAND)

Reference Authentication Navigation Data (RAND) is generated from Mask Navigation Message (MNAV) as shown in Figure 5-31. Key ID, SALT data and Reserve bits are added to Mask Navigation Message (MNAV) to generate Reference Authentication Navigation Data (RAND) as shown in Figure 5-31.

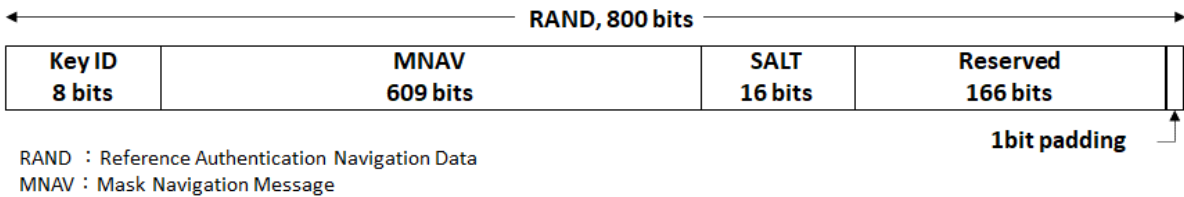


Figure 5-31: Format of QZSS CNAV2 RAND

Table 5-11: Parameter Definitions of QZSS CNAV2 RAND

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
Key ID	ID of a public key used to verify a digital signature. See 4.5	8	1	108	MSB is fixed to zero Only the 7 LSBs are used
MNAV	Output of AND operation between Reference Navigation Message and Mask Data as shown in Figure 5-30:	609	—	—	
SALT	A random value generated from true random numbers.	16	—	—	A different value is set each time a digital signature is generated.
Reserved	Reserved Bit	172	—	—	All bits are set to zero.

5.3.6 Hash Message Digest (HASH)

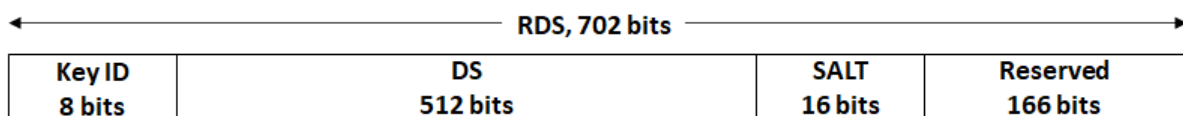
Hash message digest is computed using a hash function based on SHA-256.
See 7.2 Hash Method for details on Hash function and Hash message digest.

5.3.7 Digital Signature (DS)

Digital signature is generated by using ECDSA algorithm. See 7.1 Digital Signature and 7.1.2 Polynomial Curve for details on algorithms related with digital signature generation.

5.3.8 CNAV2 Reformat Digital Signature (RDS)

The 512-bit Digital Signature (DS) data is reformatted as shown in Figure 5-32. 8-bit Key ID, 16-bit SALT and 166 reserve bits are added to 512-bit digital signature data. The total size of reformatted digital signature data is 702 bits. Table 5-12 describes QZSS CNAV2 RDS parameters.



RDS : Reformat Digital Signature
DS : Digital Signature

Figure 5-32: Format of QZSS CNAV2 RDS

Table 5-12: Parameter Definitions of QZSS CNAV2 RDS

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
Key ID	ID of a public key used to verify a digital signature. See 4.5	8	1	108	MSB is fixed to zero Only the 7 LSBs are used Same as in Table 5-11:
DS	Digital signature data generated with the private key corresponding to the KEY ID based on the reference data	512	—	—	
SALT	A random value generated from true random numbers.	16	—	—	A different value is set each time a digital signature is generated.
Reserved	Reserve Bit	166	—	—	All bits are set to zero.

5.3.9 CNAV2 Digital Signature Segment (DSS)

The 702 bits of Reformat Digital Signature (RDS) message is split into three segments of each 234 bits. A 2-bit Segment ID is added to each segment which makes each segment 236 bits. Figure 5-33 shows the format of each Digital Signature Segment (DSS).

Table 5-13 describes the DSS parameters of CNAV2.

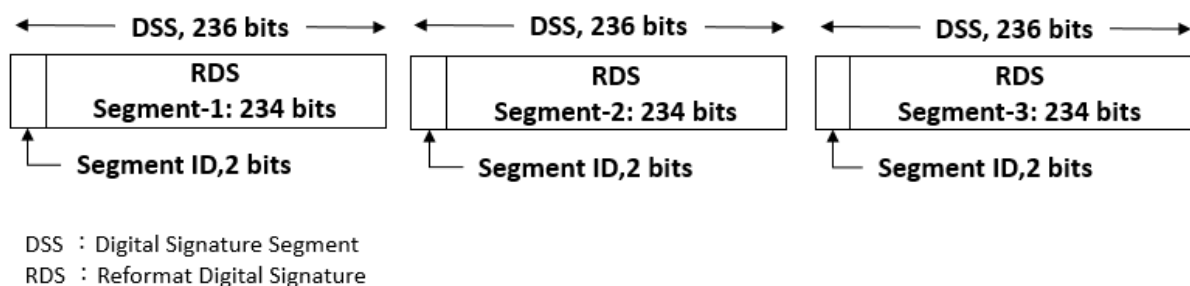


Figure 5-33: Format of QZSS CNAV2 DSS

Table 5-13: Parameter Description of QZSS CNAV2 DSS

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
Segment ID	ID to indicate the corresponding segment data	2	1	3	00b: Unused 01b: Segment 1 10b: segment 2 11b: Segment 3
RDS	Digital Signature data split into three segments	234	—	—	

5.3.10 CNAV2 Navigation Message Authentication Frame (NMAF)

The Digital Signature Segment data shown in Figure 5-33 are inserted into the navigation message of QZSS CNAV2 sub-frame 3 as shown in Figure 5-34.

Page ID indicates whether the sub-frame has a digital signature segment data in the CNAV2 message. Page ID 60 indicates that the message has a digital signature for signal authentication. Table 5-4 describes QZSS CNAV2 NMAF parameters.

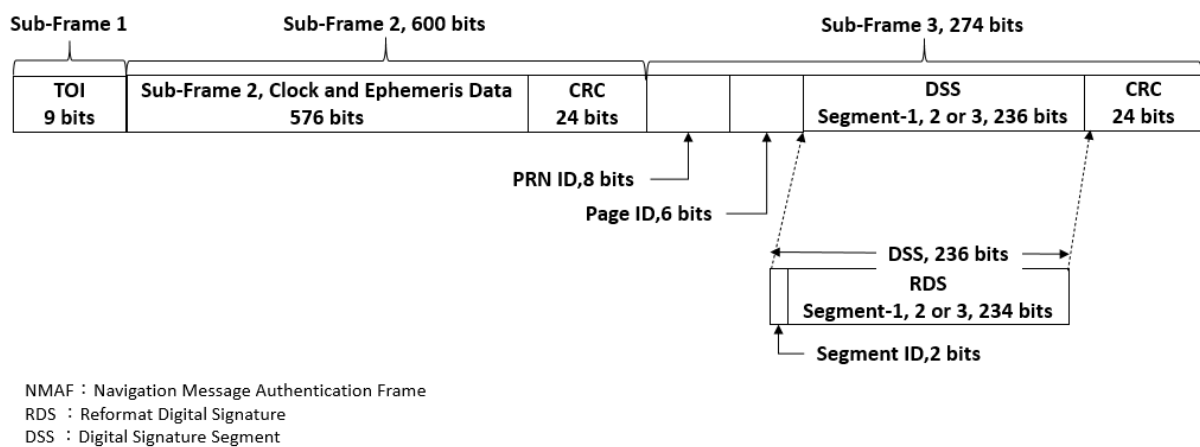


Figure 5-34: Format of QZSS CNAV2 NMAF

Table 5-14: Description of QZSS CNAV2 NMAF parameters

Parameter	Description	Length [bits]	Effective Range		Default Value	Remarks
			Min	Max		
TOI	TOI data as described in CNAV2 message	9	—	—	—	Sub-frame 1 data
Sub-frame 2 data	Sub-frame 2 data as described in CNAV2 message	600	—	—	—	Clock and Ephemeris related data
PRN ID	PRN ID as described in CNAV2 message	8	—	—	—	
Page ID	Page ID as described in CNAV2 message	6	—	—	60	Page ID 60 indicates frame with digital signature data for authentication
DSS	DSS data split into three segments	236	—	—	—	
CRC	CRC of Sub-frame 3 as described in CNAV2 message	24	—	—	—	

5.3.11 CNAV2 NMAF Broadcast Pattern

Table 5-15 shows QZSS L1C navigation message broadcast pattern for sub-frames 1 to 3. Sub-frame 3 broadcasts Digital Signature Segment data by replacing Ionosphere and UTC parameters in CNAV2 as shown in Figure 5-37.

Table 5-15 shows the broadcast pattern of one Digital Signature data (QZSS CNAV2 NMAF 1/3 to 3/3). The broadcast of one digital signature (3 NMAF) takes 54 seconds and occurs 15 times (45 NMAF) every hour. RNAV shall be SF1, SF2 and SF3, which are generally broadcast following the last NMAF.

To authenticate one of the CNAV2 signals, it is necessary to receive all three digital signature segments. It takes at least 54 seconds to get the digital signature data of one CNAV message of the QZSS L1C signal. However, it can take up to 270 seconds to retrieve the next digital signature data. This additional time of 216 seconds is because the page 1,2,3,4,61 broadcast takes place after the broadcast of each digital signature of the CNAV2 message is completed.

Table 5-15: CNAV2 NMAF Broadcast Pattern

Time [sec]	Subframe ID	Subframe ID	Subframe ID
T	Subframe 1	Subframe 2	Subframe 3 Page ID 60, NMAF (1/3)
T+18	Subframe 1	Subframe 2	Subframe 3 Page ID 60, NMAF (2/3)
T+36	Subframe 1	Subframe 2	Subframe 3 Page ID 60, NMAF (3/3)
T+54	Subframe 1	Subframe 2	Subframe 3

T=0, 288, 576, 720, 1008, 1296, 1440, 1728, 2016, 2160, 2448, 2736, 2880, 3168, 3456 (every hour)

* : Shading represents Navigation Message used for RNAV

6 GNSS Navigation Message Authentication

QZSS provides GNSS navigation message authentication for GPS LNAV, CNAV, CNAV2 messages and Galileo I/NAV, F/NAV messages using QZSS L6E signal. See Table 3-2

6.1 GPS LNAV Message Authentication

6.1.1 Overview

Figure 6-1 shows overview of GPS LNAV authentication message generation process. RNAV is generated from sub-frames 1, 2 and 3 of LNAV. The RTOW in RNAV is generated from TOW in LNAV. Reference TOW (RTOW) (Refer 6.6) explains how RTOW and TOW are handled while generating RNAV. RAND message is generated as shown in Figure 6-9. HASH is computed on this RAND message and signed using the private key to generate DS data. The reformat DS data (RDS) is broadcasted using QZSS L6E signal in QZNMA message. The broadcast period and frequency are explained in 6.7 GNSS Digital Signature Generation Frequency.

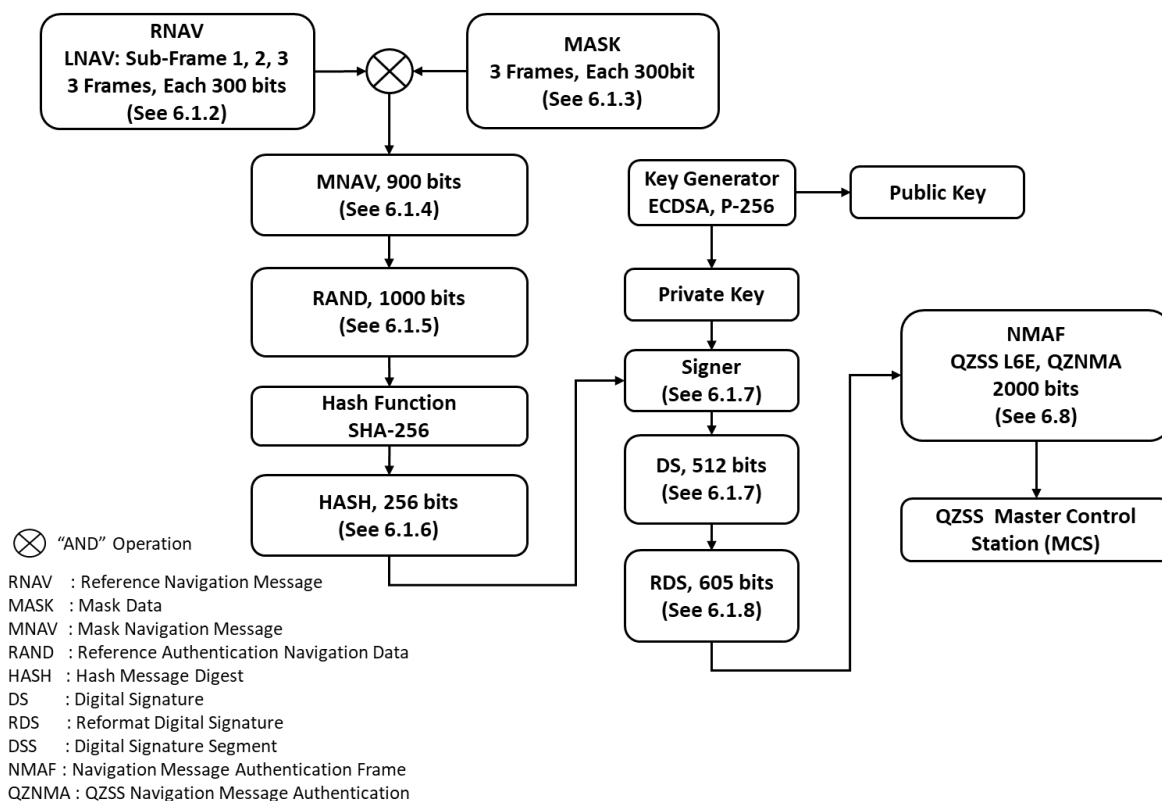


Figure 6-1: GPS LNAV authentication message generation process

6.1.2 GPS LNAV Reference Navigation Message (RNAV)

Figure 6-2, Figure 6-3 and Figure 6-4 show GPS LNAV messages for sub-frame 1, 2 and 3 that are used to generate RNAV. The TOW in LNAV is modified to generate RTOW in RNAV. See 6.6, Reference TOW (RTOW) for more information on how RTOW is generated.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30		
WORD1	PREAMBLE								TLM Message														ISF	R	Parity							
WORD2	TOW-Count Message																	Alert	AS	SubFrameID	PC	Parity										
WORD3	WN(WeekNumber)										C/A OR P ON L2			URA INDEX				SV HEALTH					IODC(2 MSBs)		Parity							
WORD4	L2P	Reserved																									Parity					
WORD5	Reserved																													Parity		
WORD6	Reserved																													Parity		
WORD7	Reserved																TGD(Group delay between SV clock and L1C/A)														Parity	
WORD8	IODC(8 LSBs)								Toc(Epoch of SV clock)																					Parity		
WORD9	af2(SV clock drift rate)								af1(SV clock drift)																						Parity	
WORD10	af0(SV clock bias)																						PC		Parity							

Figure 6-2: GPS LNAV Sub-frame 1 navigation message

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30					
WORD1	PREAMBLE								TLM Message														ISF	R	Parity										
WORD2	TOW-Count Message																	Alert	AS	SubFrameID	PC	Parity													
WORD3	IODE								Crs																						Parity				
WORD4	Δn																	M0(8 MSBs)													Parity				
WORD5	M0(24 LSBs)																															Parity			
WORD6	Cuc																	e(8 MSBs)													Parity				
WORD7	e(24 LSBs)																															Parity			
WORD8	Cus																	$\sqrt{A}$ (8 MSBs)													Parity				
WORD9	$\sqrt{A}$ (24 LSBs)																															Parity			
WORD10	toc																	Fit	AODO				PC	Parity											

Figure 6-3: GPS LNAV Sub-frame 2 navigation message

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30						
WORD1	PREAMBLE								TLM Message														ISF	R	Parity											
WORD2	TOW-Count Message																	Alert	AS	SubFrameID	PC		Parity													
WORD3	C_{ic}																$\Omega_0(8\text{ MSBs})$												Parity							
WORD4	$\Omega_0(24\text{ LSBs})$																														Parity					
WORD5	C_{is}																$i_0(8\text{ MSBs})$												Parity							
WORD6	$i_0(24\text{ LSBs})$																														Parity					
WORD7	C_{rc}																$\omega(8\text{ MSBs})$												Parity							
WORD8	$\omega(24\text{ LSBs})$																														Parity					
WORD9	Ω																														Parity					
WORD10	IODE								IDOT														PC		Parity											

Figure 6-4: GPS LNAV Sub-frame 3 navigation message

6.1.3 GPS LNAV Mask Data (MASK)

Figure 6-5, Figure 6-6 and Figure 6-7 show Mask Data of GPS LNAV sub-frame 1, sub-frame 2 and sub-frame 3 respectively. Mask Data is used to perform “AND” operation with RNAV to select only required data bits from GPS LNAV message. The Mask Data has “1” where RNAV data bits are used. The Mask Data has “0” where RNAV data bits are not used.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
WORD1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0
WORD2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
WORD3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD10	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0

Figure 6-5: MASK for GPS LNAV Sub-frame 1

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
WORD1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0
WORD2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
WORD3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD10	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0

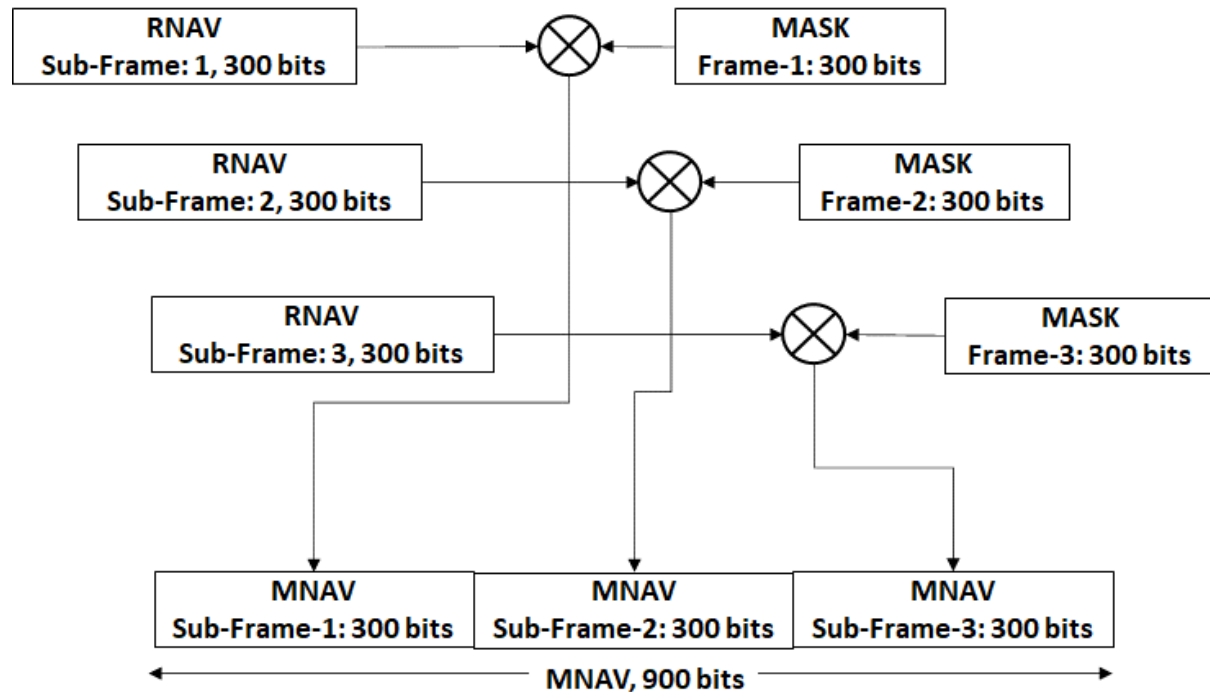
Figure 6-6: MASK for GPS LNAV Sub-frame 2

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
WORD1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0
WORD2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
WORD3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0
WORD10	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0

Figure 6-7: MASK for GPS LNAV Sub-frame 3

6.1.4 GPS LNAV Mask Navigation Message (MNAV)

Figure 6-8 shows GPS LNAV MNAV generation method from RNAV and MASK data frames. An “AND” operation is performed between each RNAV frame and MASK data frame to generate MNAV data frames. Since there are three RNAV frames of each 300 bits, the total size of MNAV frame is 900 bits.



⊗ AND Operation

RNAV : Reference Navigation Message

MASK : Mask Data

MNAV : Mask Navigation Message

Figure 6-8: GPS MNAV data generation process

6.1.5 GPS LNAV Reference Authentication Navigation Data (RAND)

Figure 6-9 shows GPS LNAV RAND message format. RAND message is generated by adding NMA ID, RTOW, SV ID, Message Type, Reference Ephemeris1, Reference Ephemeris2, Reference Clock, Key ID, MNAV, SALT and Reserved bits to generate a 1000-bit GPS LNAV RAND message from MNAV. Description of RAND message parameters are given in Table 6-1.

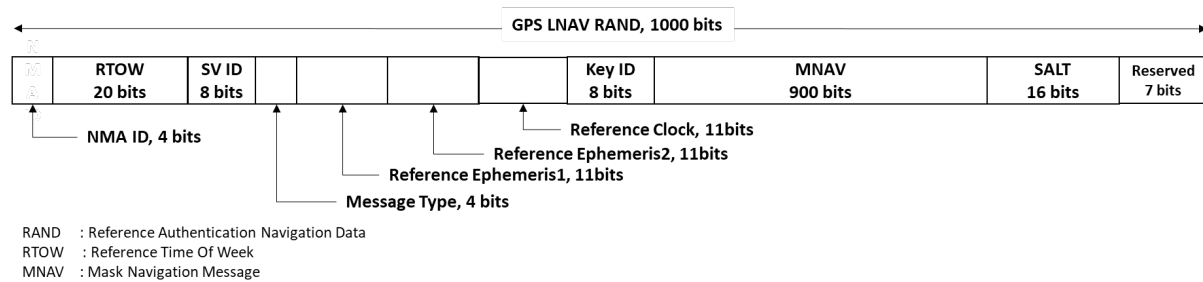


Figure 6-9: GPS LNAV RAND message format

Table 6-1: Description of Parameters for GPS LNAV RAND Message

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
NMA ID	ID to indicate that this message is for QZNMA in QZSS L6E message. NMA ID: 0000b for QZNMA If NMA ID is other than 0000b, RDS message (other than NMA ID) contains all zero. In this case, RDS data is not used for authentication.	4	—	—	
RTOW	Reference TOW RTOW refers to GPS TOW of LNAV, CNAV and CNAV2 or Galileo TOW of I/NAV and F/NAV message. GPS LNAV: 17 bits of TOW are assigned to 17 LSBs of RTOW 3 MSBs of RTOW are assigned "0" GPS CNAV: 17 bits of TOW are assigned to 17 LSBs of RTOW 3 MSBs of RTOW are assigned "0" GPS CNAV2: 8 bits of ITOW are assigned to 8 LSBs of RTOW, 9 bits of TOI are assigned to the next 9 LSBs of RTOW, 3 MSBs of RTOW are assigned "0"	20	—	—	RTOW = TOW + X X = 0 or 5 minutes RTOW may be the same as TOW or it may have an offset of X minutes See 6.6 Reference TOW (RTOW)
SV ID	ID to indicate satellite type and PRN ID of each satellite.	8	1	63	GPS SV ID
			65	127	Galileo SV ID
			129	191	SBAS SV ID (Future use)
			193	202	QZSS SV ID (Future use)

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
Message Type	Identify message type of GPS or Galileo signals GPS LNAV : 0001b GPS CNAV : 0010b GPS CNAV2 : 0011b Galileo F/NAV : 0100b Galileo I/NAV : 0101b	4	1	15	0000b is not used
Reference Ephemeris1	Reference to Ephemeris time related information of GPS and Galileo GPS LNAV : 8bits of IODE of Subframe 2 (Ephemeris 1) GPS CNAV : 11bits of toe of Message Type 10 (Ephemeris 1) GPS CNAV2 : 11bits of toe of Subframe 2 Galileo F/NAV : 10bits of IODnav of Page Type 1 Galileo I/NAV : 10bits of IODnav of Word Type 1 (Ephemeris 1/4)	11	—	—	GPS LNAV : Only the 8 LSB are used. 3 bits of the MSB is fixed to zero. Galileo F/NAV : Galileo I/NAV : Only the 10 LSB are used. 1 bit of the MSB is fixed to zero.
Reference Ephemeris2	Reference to Ephemeris time related information of GPS and Galileo GPS LNAV : 8bits of IODE of Subframe 3 (Ephemeris 2) GPS CNAV : 11bits of toe of Message Type 11 (Ephemeris 2) GPS CNAV2 : all zero Galileo F/NAV : all zero Galileo I/NAV : all zero	11	—	—	GPS LNAV : Only the 8 LSB are used. 3 bits of the MSB is fixed to zero.
Reference Clock	Reference to Clock time related information of GPS and Galileo GPS LNAV : 10bits of IODC of Subframe 1 (SV Clock) GPS CNAV : 11bits of toc of Message Type 30 to 37 (SV Clock Parameters) GPS CNAV2 : all zero Galileo F/NAV : all zero Galileo I/NAV : all zero	11	—	—	GPS LNAV : Only the 10 LSB are used. 1 bit of the MSB is fixed to zero.
Key ID	ID of a public key used to verify a digital signature, defined in section 4.5	8	1	108	Only the 7 LSB are used. Bit 1 of the MSB is fixed to zero.
MNAV	Output of AND operation between RNAV and MASK	900	—	—	See Figure 6-8

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
SALT	A random value generated from true random numbers.	16	—	—	
Reserved	Reserved Bit	7	—	—	All bits are set to zero

6.1.6 Hash Message Digest (HASH)

Hash message digest is computed using a hash function based on SHA-256.
See 7.2 Hash Method for details on Hash function and Hash message digest.

6.1.7 Digital Signature (DS)

Digital signature is generated by using ECDSA algorithm. See 7.1 Digital Signature and 7.1.2 Polynomial Curve for details on algorithms related with digital signature generation.

6.1.8 GPS LNAV Reformat Digital Signature (RDS)

The RDS format of GPS LNAV message digital signature data is shown Figure 6-10. 4-bit NMA ID, 20-bit RTOW, 8-bit SV ID, 4-bit Message Type, 11-bit Reference Ephemeris1, 11-bit Reference Ephemeris2, 11-bit Reference Clock, 8-bit Key ID, 16-bit SALT are added to DS data to generate RDS message format. Description of these parameters is given in Table 6-2.

The format of RDS for all GPS and Galileo signals are the same as shown in Figure 6-10 and as described in Table 6-2.

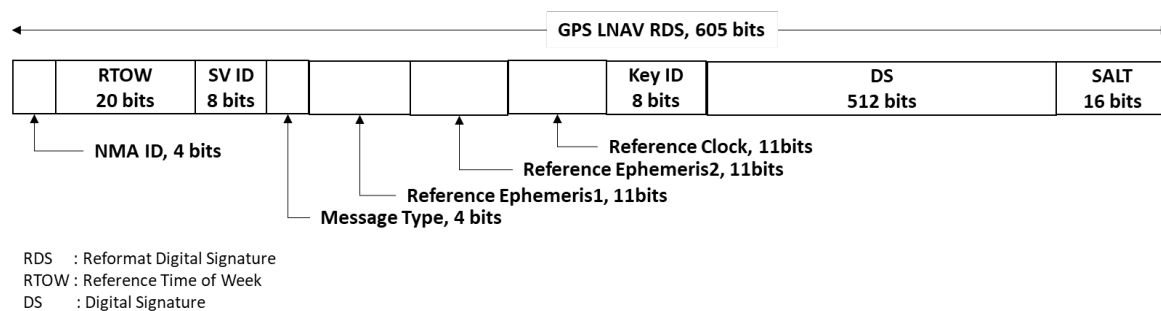


Figure 6-10: GPS LNAV RDS message format

Table 6-2: Parameter Description of GPS LNAV RDS Message

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
NMA ID	See Table 6-1	4	—	—	

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
RTOW	See Table 6-1	20	—	—	
SV ID	See Table 6-1	8	—	—	
Message Type	See Table 6-1	4	—	—	
Reference Ephemeris1	See Table 6-1	11	—	—	
Reference Ephemeris2	See Table 6-1	11	—	—	
Reference Clock	See Table 6-1	11	—	—	
Key ID	See Table 6-1	8	1	108	
Digital Signature	Digital signature data generated with the private key corresponding to the Key ID based on RAND	512	—	—	See 7.1 Digital Signature
SALT	See Table 6-1	16	—	—	

The GPS LNAV RDS data is broadcasted using QZSS L6E signal by embedding it into QZSS NMA frame as shown in Figure 6-52. Each QZSS NMA frame can accommodate up to two RDS data. See 6.9 QZSS L6E QZNMA Broadcast Pattern for details on broadcasting of GPS LNAV RDS data.

6.2 GPS CNAV Message Authentication

6.2.1 Overview

Figure 6-11 shows overview of GPS CNAV authentication message generation process. RNAV is generated from CNAV message types 10, 11 and one of the message types that is 30 or above. The RTOW in RNAV is generated from TOW in CNAV. Reference TOW (RTOW) (See 6.6) explains how RTOW and TOW are handled while generating RNAV. RAND message is generated as shown in Figure 6-19. HASH is computed on this RAND message and signed using the private key to generate DS data. The reformat DS data (RDS) is broadcasted using QZSS L6E signal in QZNMA message. The broadcast period and frequency are explained in 6.7 GNSS Digital Signature Generation Frequency.

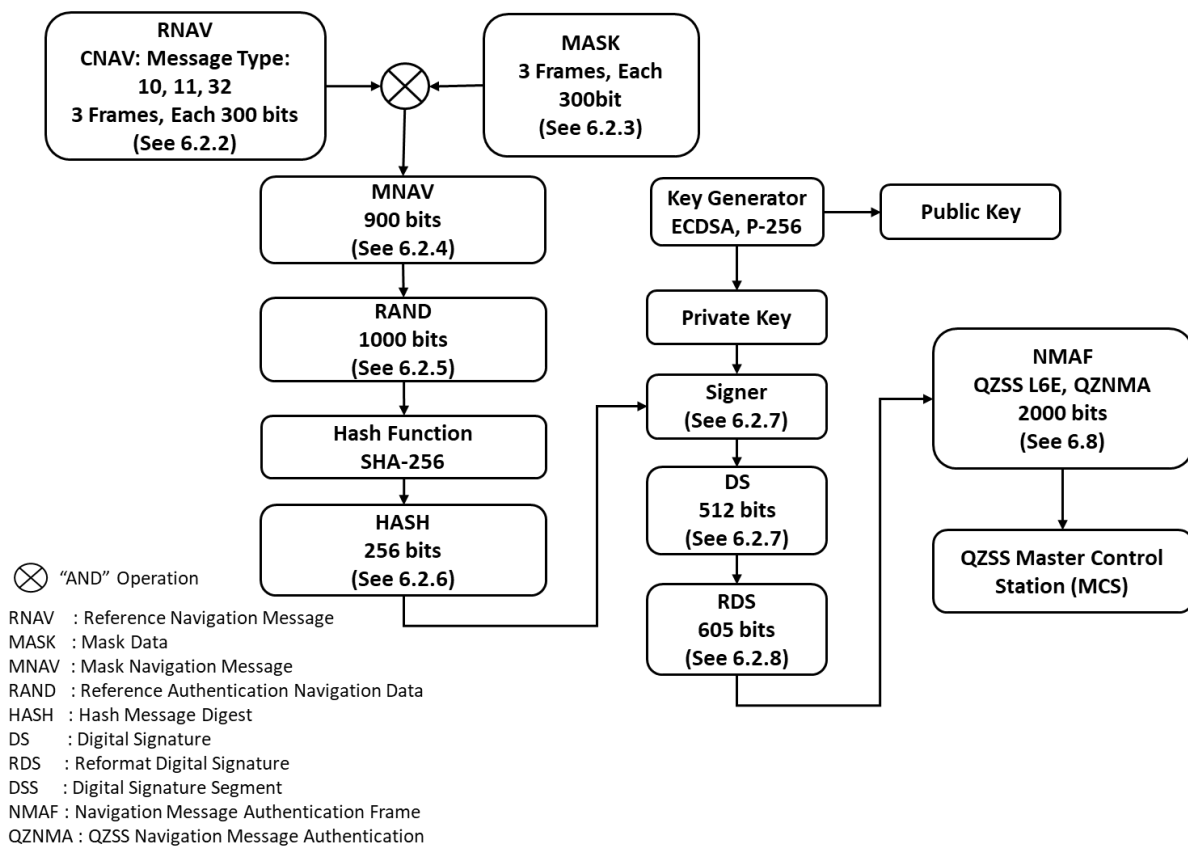


Figure 6-11: GPS CNAV authentication message generation process

6.2.2 GPS CNAV Reference Navigation Message (RNAV)

Figure 6-12, Figure 6-13 and Figure 6-14 show GPS CNAV messages for sub-frame 1, 2 and 3 that are used to generate RNAV. The TOW in CNAV is modified to generate RTOW in RNAV. See 6.6, Reference TOW (RTOW) for more information on how RTOW is generated.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30						
1	PREAMBLE(8bit)								PRN(6bit)						Message Type ID(6bit)						Message TOW Count(10 MSBs)															
2	Message TOW Count(7 LSBs)								WNn(13bit)													L1 Health			L2 Health			L5 Health			top(6 MSBs)					
3	top (5 LSBs)					URAed index(5bit)					toe(11bit)										Δ A(9 MSBs)															
4	Δ A (17 LSBs)																	Ā(13 MSBs)																		
5	Ā(12 LSBs)										Δ n0(17bit)																				Reserved					
6	Δ n0(23bit)																						M0-n(8 MSBs)													
7	M0-n(25 LSBs)																									en(5 MSBs)										
8	en(28 LSBs)																												ωn(2MSBs)							
9	ωn(30 bit)																																			
10	ωn(2LSBs)		Reserved		Reserved(3bit)			CRC(24bit)																												

Figure 6-12: GPS CNAV Sub-frame 1 navigation message

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
1	PREAMBLE(8bit)								PRN(6bit)						Message Type ID(6bit)						Message TOW Count(10 MSBs)										
2	Message TOW Count(7 LSBs)							Align(DMT)	toe(11bit)											Ω_0 -n(11 MSBs)											
3	Ω_0 -n(22 LSBs)																						io-n(8 MSBs)								
4	io-n(25 LSBs)																										$\Delta \Omega$ (5 MSBs)				
5	$\Delta \Omega$ (12 LSBs)												io-n_DOT(15bit)																Cis-n(3 MSBs)		
6	Cis-n(13 LSBs)											Cic-n(16bit)																			Align(13 MSBs)
7	Crs-n(23 LSBs)																							Crc-n(7 MSBs)							
8	Crc-n(17 LSBs)																	Cus-n(13 MSBs)													
9	Cus-n(8 LSBs)								Cuc-n(21bit)																						Align(21 MSBs)
10	Reserved(6 LSBs)							CRC(24bit)																							

Figure 6-13: GPS CNAV Sub-frame 2 navigation message

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30			
1	PREAMBLE(8bit)							PRN(6bit)							Message Type ID(6bit)					Message TOW Count(10 MSBs)													
2	Message TOW Count(7 LSBs)							ARMED0	top(11bit)											URANED0 INDEX(5bit)					URANED1 INDEX(3bit)			URANED2 INDEX(3bit)					
3	toc(11bit)										af0-n(19 MSBs)																						
4	af0-n(7 LSBs)							af1-n(20bit)																							af2-n(3 MSBs)		
5	af2-n(7 LSBs)							tEOP(16bit)													PM-X(7 MSBs)												
6	PM-X(14 LSBs)														PM-dotX(15bit)																PM-dotY(10)		
7	PM-Y(20 LSBs)																										PM-dotY(10 MSBs)						
8	PM-dotY(5 LSBs)					ΔUT_1 (25 MSBs)																											
9	ΔUT_1 (6 LSBs)						$\Delta UdotT_1$ (19bit)																			Reserved(5 MSBs)							
10	Reserved(6 LSBs)							CRC(24bit)																									

Figure 6-14: GPS CNAV Sub-frame 3 navigation message

6.2.3 GPS CNAV Mask Data (MASK)

Figure 6-15, Figure 6-16 and Figure 6-17 show Mask Data of GPS CNAV sub-frame 1, sub-frame 2 and sub-frame 3 respectively. Mask Data is used to perform “AND” operation with RNAV to select only required data bits from GPS CNAV message. The Mask Data has “1” where RNAV data bits are used. The Mask Data has “0” where RNAV data bits are not used.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
10	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 6-15: MASK for GPS CNAV Sub-frame 1

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
10	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 6-16: MASK for GPS CNAV Sub-frame 2

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 6-17: MASK for GPS CNAV Sub-frame 3

6.2.4 GPS CNAV Mask Navigation Message (MNAV)

Figure 6-18 shows GPS CNAV MNAV generation method from RNAV and MASK data frames. An “AND” operation is performed between each RNAV frame and MASK data frame to generate MNAV data frames. Since there are three RNAV frames of each 300 bits, the total size of MNAV frame is 900 bits.

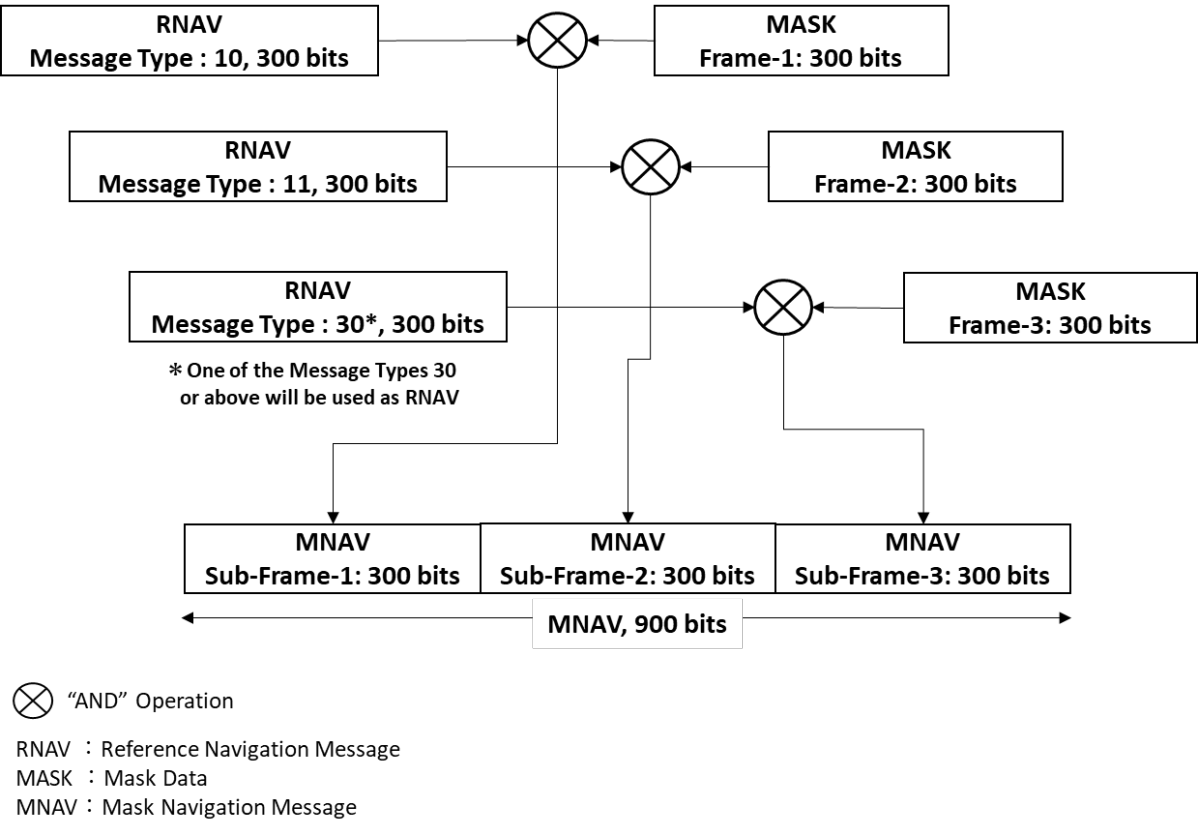


Figure 6-18: GPS MNAV data generation process

6.2.5 GPS CNAV Reference Authentication Navigation Data (RAND)

Figure 6-19 shows GPS CNAV RAND message format. RAND message is generated by adding NMA ID, RTOW, SV ID, Message Type, Reference Ephemeris1, Reference Ephemeris2, Reference Clock, Key ID, MNAV, SALT and Reserved bits to generate a 1000-bit GPS CNAV RAND message from MNAV. Description of RAND message parameters are given in Table 6-3.

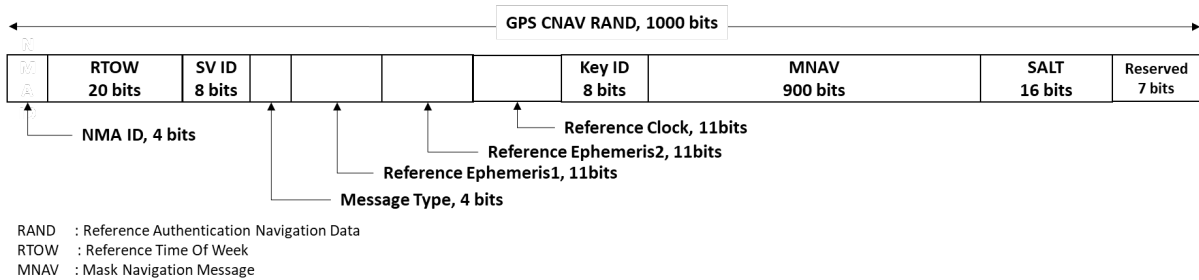


Figure 6-19: GPS CNAV RAND Format

Table 6-3: Parameter Definitions of GPS CNAV RAND

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
NMA ID	See Table 6-1	4	—	—	
RTOW	See Table 6-1	20	—	—	
SV ID	See Table 6-1	8	—	—	
Message Type	See Table 6-1	4	—	—	GPS LNAV: 0001b GPS CNAV: 0010b GPS CNAV2: 0011b Galileo F/NAV: 0100b Galileo I /NAV: 0101b
Reference Ephemeris1	See Table 6-1	11	—	—	
Reference Ephemeris2	See Table 6-1	11	—	—	
Reference Clock	See Table 6-1	11	—	—	
Key ID	See Table 6-1	8	1	108	
MNAV	Output of AND operation between RNAV and MASK as shown Figure 6-18	900	—	—	
SALT	See Table 6-1	16	—	—	
Reserved	Reserved Bit	7	—	—	All bits are set to zero.

6.2.6 Hash Message Digest (HASH)

Hash message digest is computed using a hash function based on SHA-256.

See 7.2 Hash Method for details on Hash function and Hash message digest.

6.2.7 Digital Signature (DS)

Digital signature is generated by using ECDSA algorithm. See 7.1 Digital Signature and 7.1.2 Polynomial Curve for details on algorithms related with digital signature generation.

6.2.8 GPS CNAV Reformat Digital Signature (RDS)

GPS CNAV RDS format is the same as Section 6.1.8, Figure 6-10 and Table 6-2.

GPS CNAV RDS data is broadcasted using QZSS L6E signal by embedding it into QZSS NMA frame as shown in Figure 6-52. Each QZSS NMA frame can accommodate up to two RDS data. See 6.9 QZSS L6E QZNMA Broadcast Pattern for details on broadcasting of GPS CNAV RDS data.

6.3 GPS CNAV2 Message Authentication

6.3.1 Overview

Figure 6-20 shows overview of GPS CNAV2 authentication message generation process. RNAV is generated from CNAV2 sub-frames 1 and 2. The RTOW in RNAV is generated from TOW in CNAV2. Reference TOW (RTOW) (See 6.6) explains how RTOW and TOW are handled while generating RNAV. RAND message is generated as shown in Figure 6-26. HASH is computed on this RAND message and signed using the private key to generate DS data. The reformat DS data (RDS) is broadcasted using QZSS L6E signal in QZNMA message. The broadcast period and frequency are explained in 6.7 GNSS Digital Signature Generation Frequency.

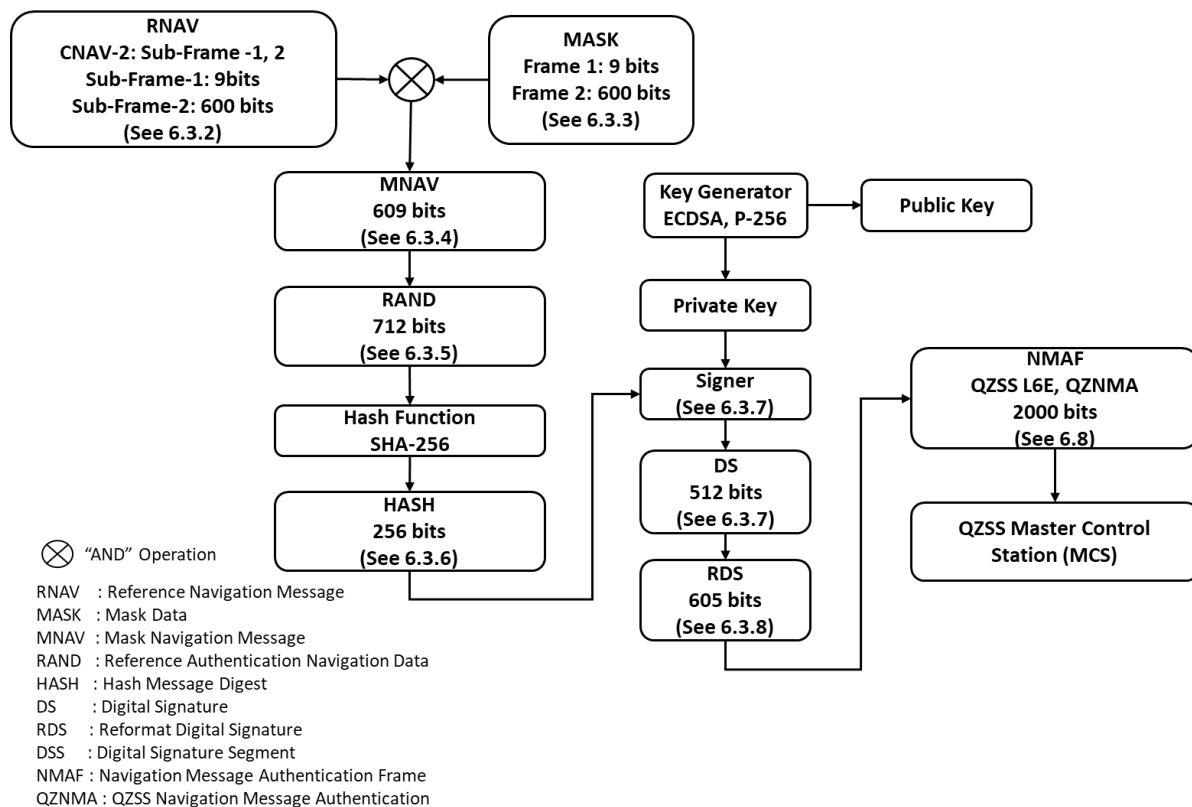


Figure 6-20: GPS CNAV2 authentication message generation process

6.3.2 GPS CNAV2 Reference Navigation Message (RNAV)

Figure 6-21 and Figure 6-22 show GPS CNAV2 messages for sub-frame 1 and 2 that are used to generate RNAV. The TOW in CNAV2 is modified to generate RTOW in RNAV. See 6.6, Reference TOW (RTOW) for more information on how RTOW is generated.

	1	2	3	4	5	6	7	8	9
1	TOI(9bit)								

Figure 6-21: GPS CNAV2 Sub-frame 1 navigation message

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30			
1	WN(13bit)													ITOW(8bit)								top(9 MSBs)											
2	top(2 LSBs)		13C INDEX		URA ED INDEX(5bit)				toe(11bit)										ΔA(11 MSBs)														
3	ΔA(15 LSBs)															dot A(15 MSBs)																	
4	dot A(10 LSBs)										Δ n0										Δ dot n0(3 MSBs)												
5	Δ dot n0(20 LSBs)																				M 0-n(10 MSBs)												
6	M 0-n(23 LSBs)																							en(7 MSBs)									
7	en(26 LSBs)																										ωn(4 MSBs)						
8	ωn(29 LSBs)																													ωn(1 MSB)			
9	Ω 0-n(30bit)																																
10	Ω 0-n(2 LSBs)		i 0-n(28 MSBs)																														
11	i 0-n(5 LSBs)				Δ dot Ω(17bit)																		i 0- dot n(8 MSBs)										
12	i 0- dot n(7 LSBs)						C is-n(16bit)																		C ic-n(7 MSBs)								
13	C ic-n(9 LSBs)								C rs-n(21 MSBs)																								
14	C rs-n(3 LSBs)			C rc-n(24bit)																											C us-n(3 MSBs)		
15	C us-n(18 LSBs)																	C uc-n(12 MSBs)															
16	C uc-n(9 LSBs)							URA NED0 INDEX(5bit)				URA NED1 INDEX(3bit)				URA NED2 INDEX(3bit)				a f0-n(10 MSBs)													
17	a f0-n(16 LSBs)																a f1-n(14 MSBs)																
18	a f1-n(6 LSBs)					a f2-n(10bit)										TGD(13bit)														TGD(1 MSB)			
19	ISC L1CP(12 LSBs)												ISC L1CD(13bit)																ISC L1CD(1 MSB)		WN OP(4 MSBs)		
20	WN OP(4 LSBs)			Reserved(2bit)		CRC(24bit)																											

Figure 6-22: GPS CNAV2 Sub-frame 2 navigation message

6.3.3 GPS CNAV2 Mask Data (MASK)

Figure 6-23 and Figure 6-24 show Mask Data of GPS CNAV2 sub-frame 1 and sub-frame 2 respectively. Mask Data is used to perform “AND” operation with RNAV to select only required data bits from GPS CNAV2 message. The Mask Data has “1” where RNAV data bits are used. The Mask Data has “0” where RNAV data bits are not used.

	1	2	3	4	5	6	7	8	9
1	1	1	1	1	1	1	1	1	1

Figure 6-23: MASK for GPS CNAV2 Sub-frame 1

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
8	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
9	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
10	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
11	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
12	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
13	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
14	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
15	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
16	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
17	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
18	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
19	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
20	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 6-24: MASK for GPS CNAV2 Sub-frame 2

6.3.4 GPS CNAV2 Mask Navigation Message (MNAV)

Figure 6-25 shows GPS CNAV2 MNAV generation method from RNAV and MASK data frames. An “AND” operation is performed between each RNAV frame and MASK data frame to generate MNAV data frames. Since RNAV sub-frame 1 is 9 bits and RNAV sub-frame 2 is 600 bits, the total size of MNAV frame is 900 bits.

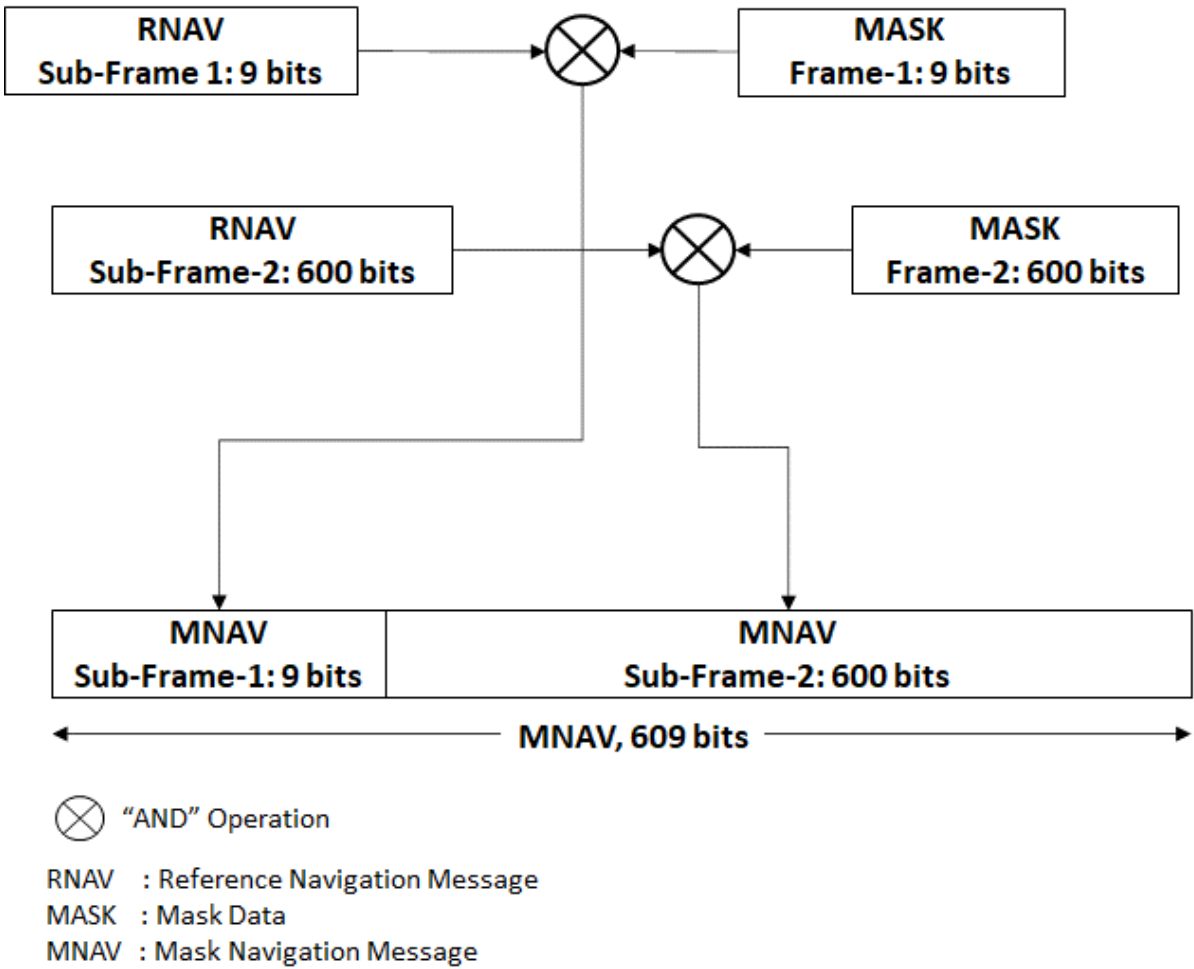


Figure 6-25: GPS MNAV data generation process

6.3.5 GPS CNAV2 Reference Authentication Navigation Data (RAND)

Figure 6-26 shows GPS CNAV2 RAND message format. RAND message is generated by adding NMA ID, RTOW, SV ID, Message Type, Reference Ephemeris1, Reference Ephemeris2, Reference Clock, Key ID, MNAV, SALT and Reserved bits to generate a 712-bit GPS CANV2 RAND message from MNAV. Description of RAND message parameters are given in Table 6-4.

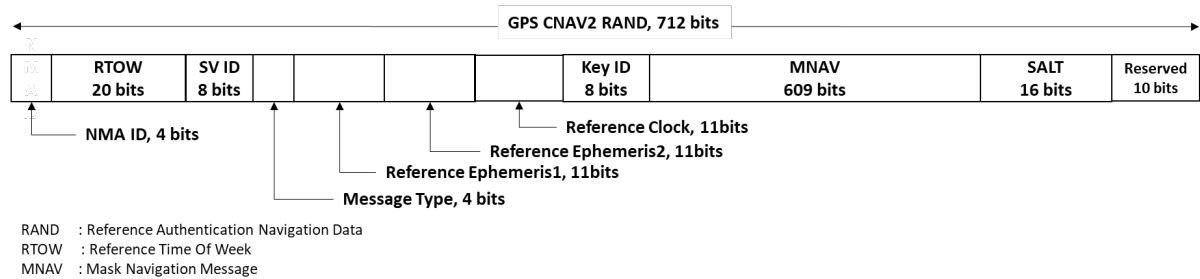


Figure 6-26: GPS CNAV2 RAND format

Table 6-4: Parameter Description of GPS CNAV2 RAND

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
NMA ID	See Table 6-1	4	—	—	
RTOW	See Table 6-1	20	—	—	
SV ID	See Table 6-1	8	—	—	
Message Type	See Table 6-1	4	—	—	GPS LNAV : 0001b GPS CNAV : 0010b GPS CNAV2 : 0011b Galileo F/NAV : 0100b Galileo I/NAV : 0101b
Reference Ephemeris1	See Table 6-1	11	—	—	
Reference Ephemeris2	See Table 6-1	11	—	—	
Reference Clock	See Table 6-1	11	—	—	
Key ID	See Table 6-1	8	1	108	
MNAV	MNAV Output of AND operation between RNAV and MASK as shown in Figure 6-25	609	—	—	
SALT	See Table 6-1	16	—	—	
Reserved	Reserved Bit	10	—	—	All bits are set to zero.

6.3.6 Hash Message Digest (HASH)

Hash message digest is computed using a hash function based on SHA-256.

See 7.2 Hash Method for details on Hash function and Hash message digest.

6.3.7 Digital Signature (DS)

Digital signature is generated by using ECDSA algorithm. See 7.1 Digital Signature and 7.1.2 Polynomial Curve for details on algorithms related with digital signature generation.

6.3.8 GPS CNAV2 Reformat Digital Signature (RDS)

GPS CNAV2 RDS format is the same as Section 6.1.8, Figure 6-10 and Table 6-2.

GPS CNAV2 RDS data is broadcasted using QZSS L6E signal by embedding it into QZSS NMA frame as shown in Figure 6-52. Each QZSS NMA frame can accommodate up to two RDS data. See 6.9 QZSS L6E QZNMA Broadcast Pattern for details on broadcasting of GPS CNAV2 RDS data.

6.4 Galileo I/NAV Message Authentication

6.4.1 Overview

Figure 6-27 shows overview of Galileo I/NAV authentication message generation process. RNAV is generated from I/NAV word type 1, 2, 3, 4 and 5. The RTOW in RNAV is generated from TOW in I/NAV. Reference TOW (RTOW) (See 6.6) explains how RTOW and TOW are handled while generating RNAV. RAND message is generated as shown in Figure 6-39. HASH is computed on this RAND message and signed using private key to generate DS data. The reformat DS data (RDS) is broadcasted using QZSS L6E signal in QZNMA message. The broadcast period and frequency are explained in 6.7 GNSS Digital Signature Generation Frequency.

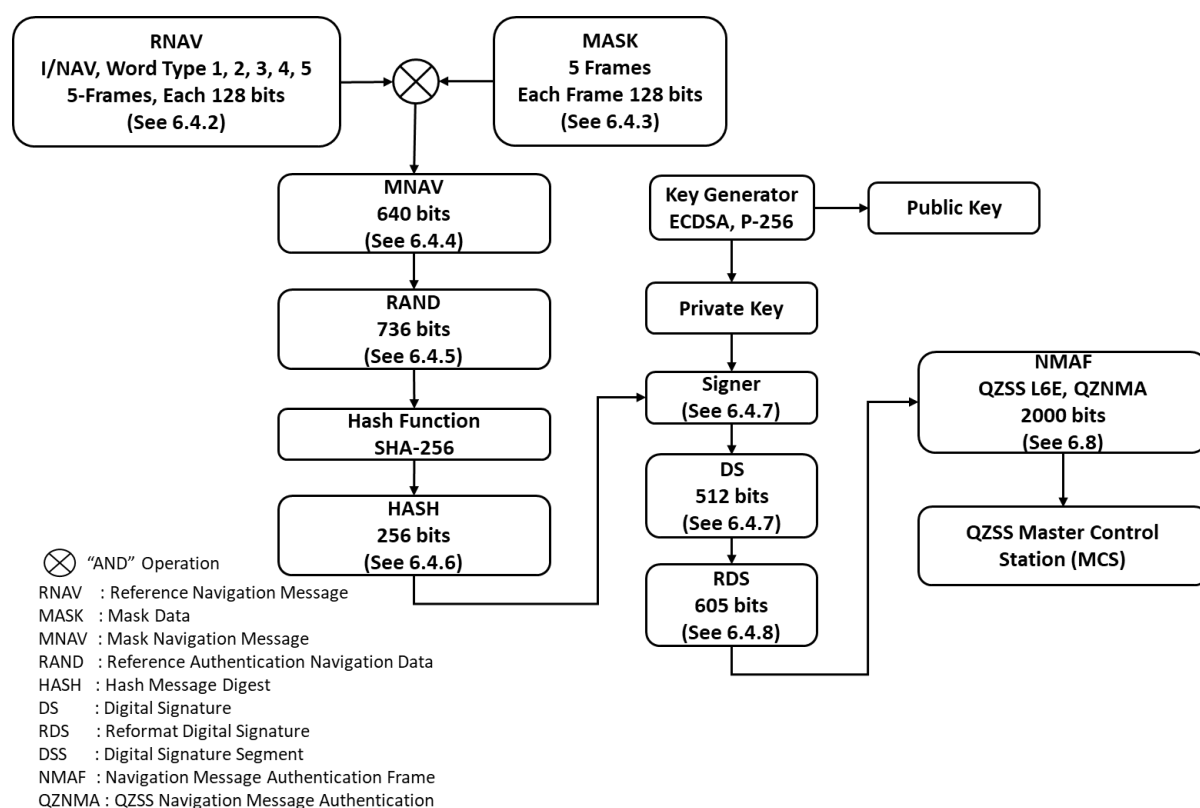


Figure 6-27: Galileo I/NAV authentication message generation process

6.4.2 Galileo I/NAV Reference Navigation Message (RNAV)

Figure 6-28, Figure 6-29, Figure 6-30, Figure 6-31 and Figure 6-32 show Galileo I/NAV messages for word type 1, 2, 3, 4 and 5 that are used to generate RNAV. The TOW in I/NAV word type 5 is used to generate RTOW in RNAV. See 6.6, Reference TOW (RTOW) for more information on how RTOW is generated.

Word Type 1: Ephemeris (1/4)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	
1	Type=1(6bit)						IODnav(10bit)										t0e(14bit)																M0(32 MSBs)
2	M0(30 LSBs)																e(2 MSBs)																
3	e(30 LSBs)																√A(2 MSBs)																
4	√A(30 LSBs)																Reserved(2bit)																

Figure 6-28: Galileo I/NAV Word Type 1 navigation message

Word Type 2: Ephemeris (2/4)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	Type=2(6bit)						IODnav(10bit)										Ω0(16 MSBs)															
2	Ω0(16 LSBs)																i0(16 MSBs)															
3	i0(16 LSBs)																ω(16 MSBs)															
4	ω(16 LSBs)																dot i(14bit)															Reserved(2bit)

Figure 6-29: Galileo I/NAV Word Type 2 navigation message

Word Type 3: Ephemeris (3/4) and SISA

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	Type=3(6bit)						IODnav(10bit)										dot Ω(16 MSBs)															
2	dot Ω(8 LSBs)								Δn(16bit)																Cuc(8 MSBs)							
3	Cuc(8 LSBs)								Cus(16bit)																Crc(8 MSBs)							
4	Crc(8LSBs)								Crs(16bit)																SISA(E1,E5b)							

Figure 6-30: Galileo I/NAV Word Type 3 navigation message

Word Type 4: SVID, Ephemeris (4/4), and Clock correction parameters

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	Type=4(6bit)					IODnav(10bit)										SVID(6bit)						Cic(10 MSBs)										
2	Cic(6 LSBs)					Cis(16bit)																t0c(10 MSBs)										
3	t0c(4 LSBs)					af0(29 MSBs)																										
4	af0(3 LSBs)					af1(21bit)																			af2(6bit)					Spare(2bit)		

Figure 6-31: Galileo I/NAV Word Type 4 navigation message

Word Type 5: Ionospheric correction, BGD, signal health and data validity status and GST

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	Type=5(6bit)						Az ai0(11bit)										Az ai1(11bit)										Az ai2(4 MSBs)					
2	Az ai2(10 LSBs)									Ionospheric disturbance flag						BGD(E1,E5a)(10bit)										BGD(E1,E5b)(7 MSBs)						
3	BGD(3 LSBs)			E5bHS			E1bHS									WN(12bit)										TOW(11 MSBs)						
4	TOW(9 LSBs)									Spare(23bit)																						

Figure 6-32: Galileo I/NAV Word Type 5 navigation message

6.4.3 Galileo INAV Mask Data (MASK)

Figure 6-33, Figure 6-34, Figure 6-35, Figure 6-36 and Figure 6-37 show Mask Data of Galileo I/NAV word type 1, 2, 3, 4, and 5 respectively. Mask Data is used to perform “AND” operation with RNAV to select only required data bits from Galileo I/NAV message. The Mask Data has “1” where RNAV data bits are used. The Mask Data has “0” where RNAV data bits are not used.

Word Type 1: Ephemeris (1/4)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Figure 6-33: MASK for Galileo FNAV Word Type 1

Word Type 2: Ephemeris (2/4)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Figure 6-34: MASK for Galileo FNAV Word Type 2

Word Type 3: Ephemeris (3/4) and SISA

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Figure 6-35: MASK for Galileo FNAV Word Type 3

Word Type 4: SVID, Ephemeris (4/4), and Clock correction parameters

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Figure 6-36: MASK for Galileo FNAV Word Type 4

Word Type 5: Ionospheric correction, BGD, signal health and data validity status and GST

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Figure 6-37: MASK for Galileo FNAV Word Type 5

6.4.4 Galileo I/NAV Mask Navigation Message (MNAV)

Figure 6-38 shows Galileo I/NAV MNAV generation method from RNAV and MASK data frames. An “AND” operation is performed between each RNAV frame and MASK data frame to generate MNAV data frames. Since there are five RNAV frames of each 128 bits, the total size of MNAV frame is 640 bits.

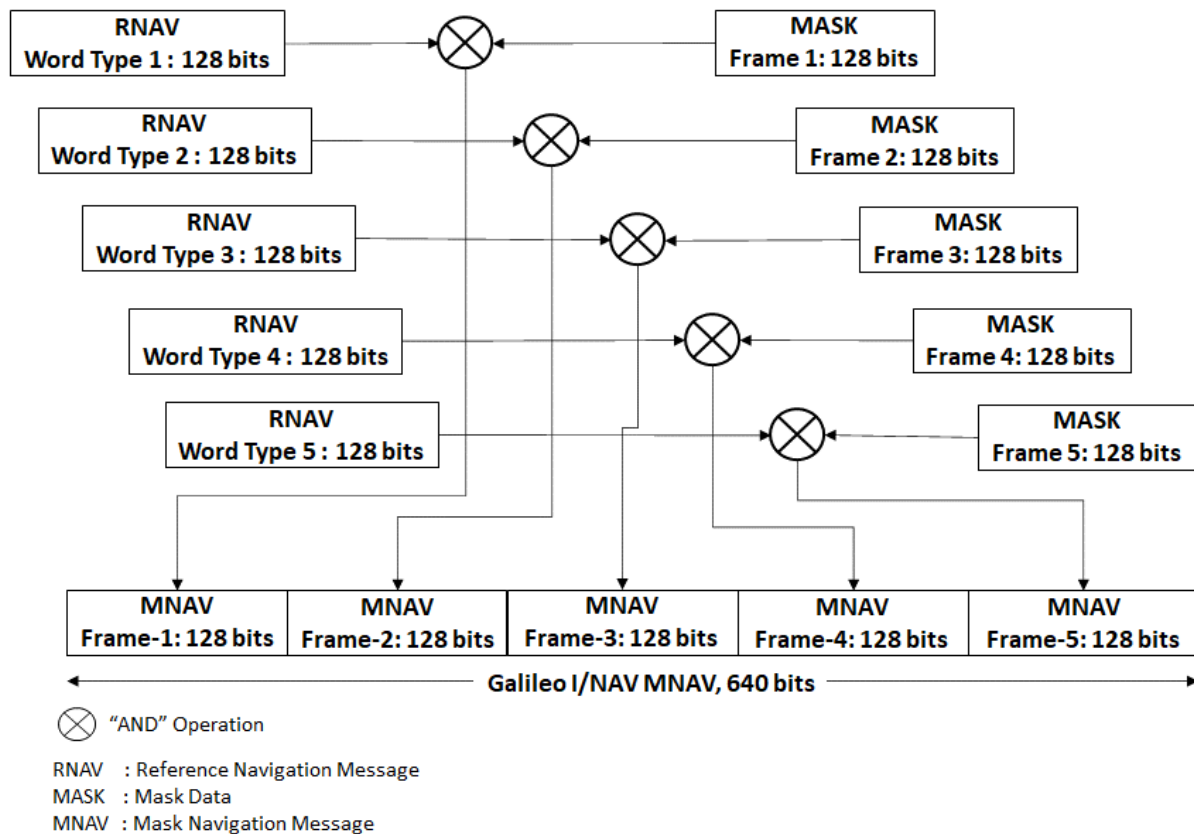


Figure 6-38: Galileo MNAV data generation process

6.4.5 Galileo I/NAV Reference Authentication Navigation Data (RAND)

Figure 6-39 shows Galileo I/NAV RAND message format. RAND message is generated by adding NMA ID, RTOW, SV ID, Message Type, Reference Ephemeris1, Reference Ephemeris2, Reference Clock, Key ID, MNAV, SALT and Reserved bits to generate a 736-bit Galileo I/NAV RAND message from MNAV. Description of RAND message parameters are given in Table 6-5.

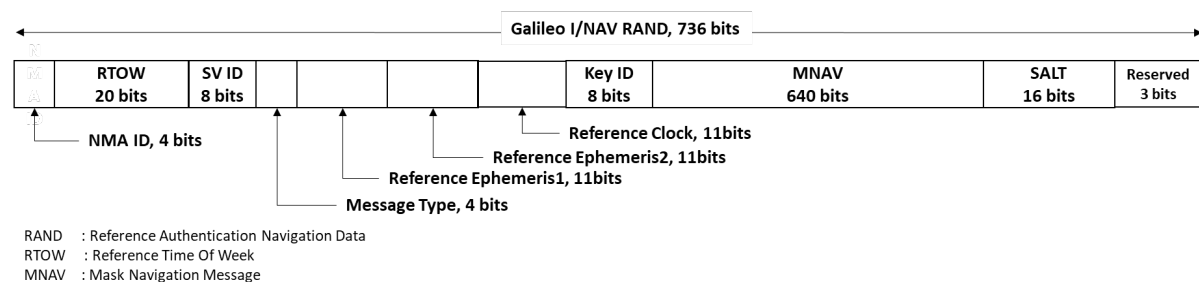


Figure 6-39: Galileo I/NAV RAND message format

Table 6-5: Parameter Description of Galileo I/NAV RAND Message

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
NMA ID	See Table 6-1	4	—	—	
RTOW	See Table 6-1	20	—	—	
SV ID	See Table 6-1	8	—	—	
Message Type	See Table 6-1	4	—	—	
Reference Ephemeris1	See Table 6-1	11	—	—	
Reference Ephemeris2	See Table 6-1	11	—	—	
Reference Clock	See Table 6-1	11	—	—	
Key ID	See Table 6-1	8	1	108	
MNAV	Output of AND operation between RNAV and MASK as shown Figure 6-38	640	—	—	
SALT	See Table 6-1	16	—	—	
Reserved	Reserved Bit	3	—	—	All bits are set to zero.

6.4.6 Hash Message Digest (HASH)

Hash message digest is computed using a hash function based on SHA-256.
See 7.2 Hash Method for details on Hash function and Hash message digest.

6.4.7 Digital Signature (DS)

Digital signature is generated by using ECDSA algorithm. See 7.1 Digital Signature and 7.1.2 Polynomial Curve for details on algorithms related with digital signature generation.

6.4.8 Galileo I/NAV Reformat Digital Signature (RDS)

Galileo I/NAV RDS format is the same as 6.1.8, Figure 6-10 and Table 6-2.
Galileo I/NAV RDS data is broadcasted using QZSS L6E signal by embedding it into QZSS NMA frame as shown in Figure 6-52. Each QZSS NMA frame can accommodate up to two RDS data. See 6.9 QZSS L6E QZNMA Broadcast Pattern for details on broadcasting of Galileo I/NAV RDS data.

6.5 Galileo F/NAV Message Authentication

6.5.1 Overview

Figure 6-40 shows overview of Galileo F/NAV authentication message generation process. RNAV is generated from F/NAV page type 1, 2, 3 and 4. The RTOW in RNAV is generated from TOW in F/NAV. Reference TOW (RTOW) (See 6.6) explains how RTOW and TOW are handled while generating RNAV. RAND message is generated as shown in Figure 6-50. HASH is computed on this RAND message and signed using the private key to generate DS data. The reformat DS data (RDS) is broadcasted using QZSS L6E signal in QZNMA message. The broadcast period and frequency are explained in 6.7 GNSS Digital Signature Generation Frequency.

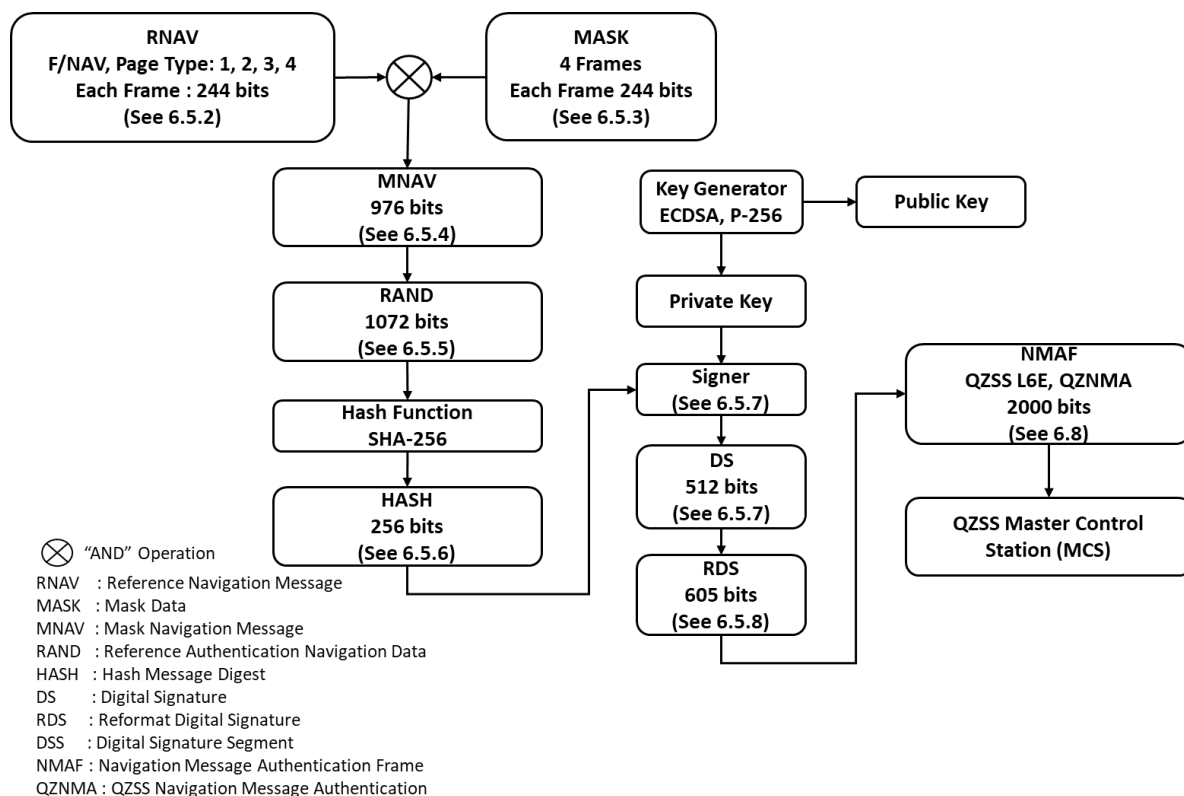


Figure 6-40: Galileo F/NAV authentication message generation process

6.5.2 Galileo F/NAV Reference Navigation Message (RNAV)

Figure 6-41, Figure 6-42, Figure 6-43 and Figure 6-44 show Galileo F/NAV messages for page type 1, 2, 3 and 4 that are used to generate RNAV. The TOW in each page type of F/NAV is used to generate RTOW in RNAV. See 6.6, Reference TOW (RTOW) for more information on how RTOW is generated.

Page Type 1: SVID, Clock correction, SISA, Ionospheric correction, BGD, GST, Signal health and Data validity status

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	
1	Type = 1(6bit)					SVID(6bit)					IODnav(10bit)										t0c(10 MSBs)												
2	t0c(4 LSBs)				af0(28 MSBs)																												
3	af0(3 LSBs)			af1(21bit)																		af2(6bit)					SISA(2 MSBs)						
4	SISA(E1,E5a)(6					ai0(11bit)										ai1(11bit)										ai2(4 MSBs)							
5	ai2(10 LSBs)								Ionospheric disturbance flag					BGD(E1,E5a)(10bit)										E5aHS			WN(5 MSBs)						
6	WN(7 LSBs)						TOW(20bit)															Spare(4 MSBs)											
7	Spare(22 LSBs)																						CRC(10 MSBs)										
8	CRC(14 LSBs)															Tail(6bit)																	

Figure 6-41: Galileo F/NAV Page Type 1 navigation message

Page Type 2: Ephemeris (1/3) and GST

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	Type = 2(6bit)						IODnav(10bit)										M0(16 MSBs)															
2	M0(16 LSBs)																dot Ω (16 MSBs)															
3	dot Ω (8 LSBs)								e(24 MSBs)																							
4	e(8 LSBs)								$\sqrt{A}$ (24 MSBs)																							
5	$\sqrt{A}$ (8 LSBs)								Ω 0(24 MSBs)																							
6	Ω 0(8 LSBs)								dot i(14bit)																WN(10 MSBs)							
7	WN(2 LSBs)		TOW(20bit)																CRC(10 MSBs)													
8	CRC(14 LSBs)														Tail(6bit)																	

Figure 6-42: Galileo F/NAV Page Type 2 navigation message

Page Type 3: Ephemeris (2/3) and GST

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	Type = 3(6bit)					IODnav(10bit)											i0(16 MSBs)															
2	i0(16 LSBs)											ω(16 MSBs)																				
3	ω(16 LSBs)											Δn(16bit)																				
4	Cuc(16bit)											Cus(16bit)																				
5	Crc(16bit)											Crs(16bit)																				
6	t0e(14bit)															WN(12bit)												TOW(6 MSBs)				
7	TOW(14 LSBs)															Spare(8bit)										CRC(10 MSBs)						
8	CRC(14 LSBs)															Tail(6bit)																

Figure 6-43: Galileo F/NAV Page Type 3 navigation message

Page Type 4: Ephemeris (3/3), GST-UTC conversion, GST-GPS conversion and TOW

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32						
1	Type = 4(6bit)						IODnav(10bit)										Cic(16bit)																					
2	Cis(16bit)																A0(16 MSBs)																					
3	A0(16 LSBs)																A1(16 MSBs)																					
4	A1(8 LSBs)								ΔtLs(8bit)								t0t(8bit)								WN0t(8bit)													
5	WN LSF(8bit)								DN(3bit)			Δt LSF(8bit)								t0G(8bit)								A0G(5 MSBs)										
6	A0G(11 LSBs)										A1G(12bit)												WN 0G(6bit)										TOW(3 MSBs)					
7	TOW(17 LSBs)																	Spare(5bit)					CRC(10 MSBs)															
8	CRC(14 LSBs)														Tail(6bit)																							

Figure 6-44: Galileo F/NAV Page Type 4 navigation message

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6.5.3 Galileo F/NAV Mask Data (MASK)

Figure 6-45, Figure 6-46, Figure 6-47 and Figure 6-48 show Mask Data of Galileo F/NAV page-type 1, 2, 3 and 4 respectively. Mask Data is used to perform “AND” operation with RNAV to select only required data bits from Galileo F/NAV message. The Mask Data has “1” where RNAV data bits are used. The Mask Data has “0” where RNAV data bits are not used.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0

Figure 6-45: MASK for Galileo F/NAV Page Type 1

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1											

Figure 6-46: MASK for Galileo F/NAV Page Type 2

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1											

Figure 6-47: MASK for Galileo F/NAV Page Type 3

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
6	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1											

Figure 6-48: MASK for Galileo F/NAV Page Type 4

6.5.4 Galileo F/NAV Mask Navigation Message (MNAV)

Figure 6-49 shows Galileo F/NAV MNAV generation method from RNAV and MASK data frames. An “AND” operation is performed between each RNAV frame and MASK data frame to generate MNAV data frames. Since there are four RNAV frames of each 244 bits, the total size of MNAV frame is 976 bits.

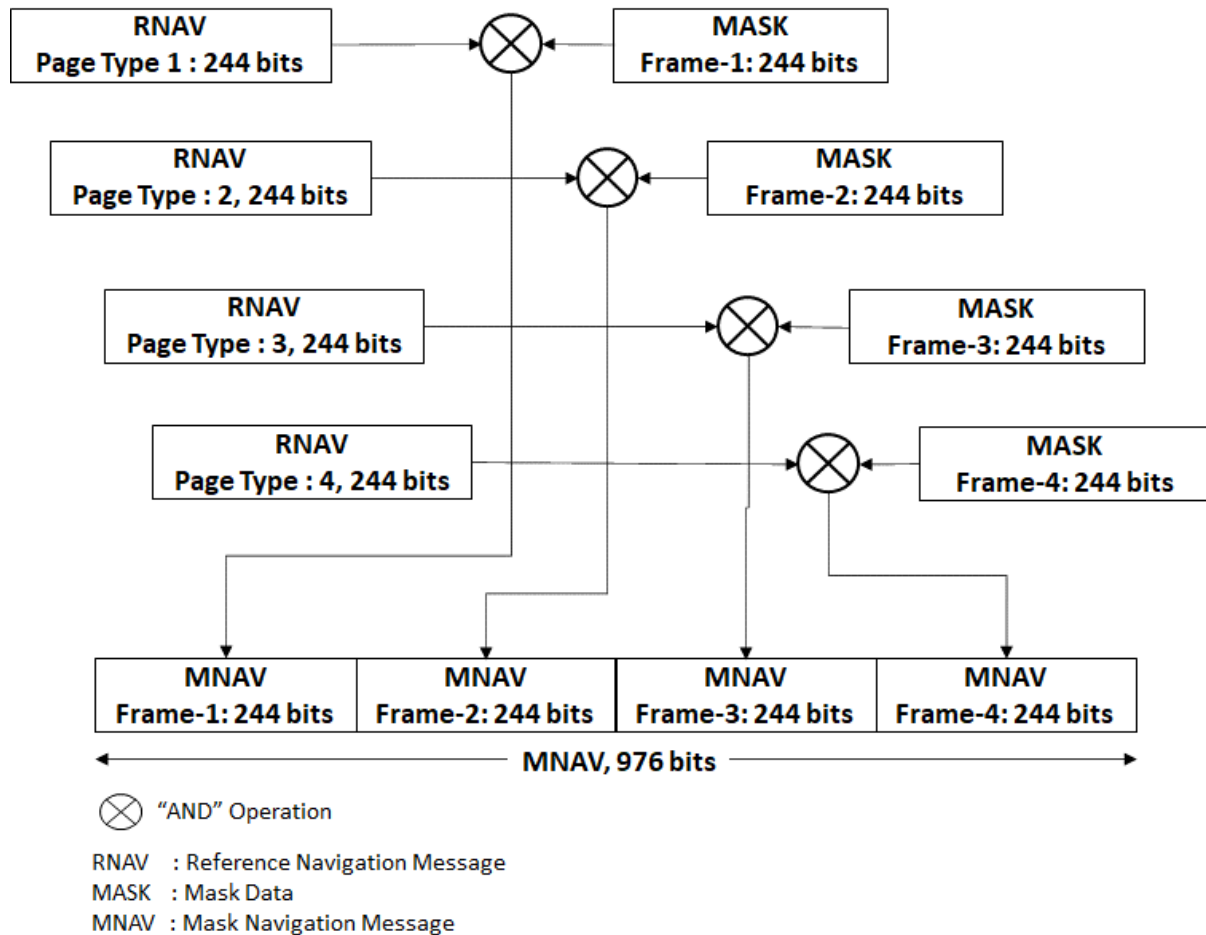


Figure 6-49: Galileo MNAV data generation process

6.5.5 Galileo F/NAV Reference Authentication Navigation Data (RAND)

Figure 6-50 shows Galileo F/NAV RAND message format. RAND message is generated by adding NMA ID, RTOW, SV ID, Message Type, Reference Ephemeris1, Reference Ephemeris2, Reference Clock, Key ID, MNAV, SALT and Reserved bits to generate a 1072-bit Galileo F/NAV RAND message from MNAV. Description of RAND message parameters are given in Table 6-6.

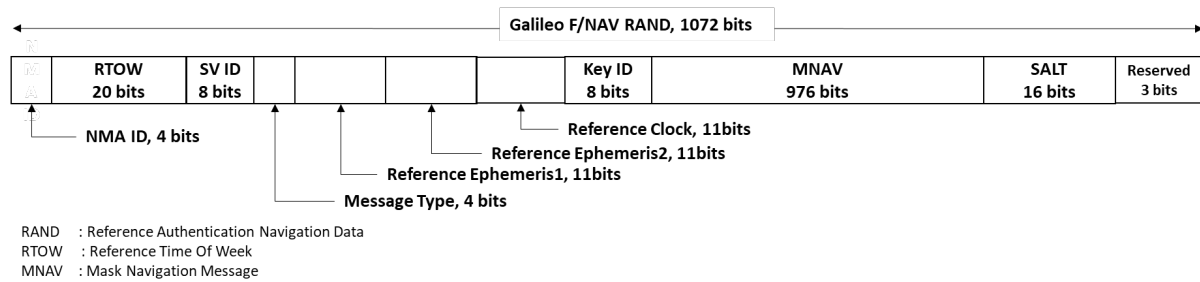


Figure 6-50: Galileo F/NAV RAND Message Format

Table 6-6: Parameter Descriptions of Galileo F/NAV RAND Message

Parameter	Description	Length [bits]	Effective Range		Remarks
			Min	Max	
NMA ID	See Table 6-1	4	—	—	
RTOW	See Table 6-1	20	—	—	
SV ID	See Table 6-1	8	—	—	
Message Type	See Table 6-1	4	—	—	
Reference Ephemeris1	See Table 6-1	11	—	—	
Reference Ephemeris2	See Table 6-1	11	—	—	
Reference Clock	See Table 6-1	11	—	—	
Key ID	See Table 6-1	8	1	108	
MNAV	Output of AND operation between RNAV and MASK as shown Figure 6-49	976	—	—	
SALT	See Table 6-1	16	—	—	
Reserved	Reserved Bit	3	—	—	All bits are set to zero.

6.5.6 Hash Message Digest (HASH)

Hash message digest is computed using a hash function based on SHA-256.
See 7.2 Hash Method for details on Hash function and Hash message digest.

6.5.7 Digital Signature (DS)

Digital signature is generated by using ECDSA algorithm. See 7.1 Digital Signature and 7.1.2 Polynomial Curve for details on algorithms related with digital signature generation.

6.5.8 Galileo F/NAV Reformat Digital Signature (RDS)

Galileo F/NAV RDS format is the same as 6.1.8, Figure 6-10 and Table 6-2.

Galileo F/NAV RDS data is broadcasted using QZSS L6E signal by embedding it into QZSS NMA frame as shown in Figure 6-52. Each QZSS NMA frame can accommodate up to two RDS data. See 6.9 QZSS L6E QZNMA Broadcast Pattern for details on broadcasting of Galileo F/NAV RDS data.

6.6 Reference TOW (RTOW)

6.6.1 RTOW Generation

TOW data in navigation message of GPS (LNAV, CNAV, CNAV2) and Galileo (I/NAV, F/NAV) are modified to RTOW by adding and offset of “X” minutes to generate RNAV for GPS (see 6.1.2, 6.2.2, 6.3.2) and Galileo (see 6.4.2, 6.5.2). Figure 6-51 shows an example of RTOW in RNAV for GPS LNAV.

Notes:

1. If $X = 0$; there are no offsets and hence the RTOW values are the same as the TOW values.
2. If $X = 5$; the RTOW values are generated by adding 5 minutes (300 seconds) to TOW values.
3. The offset value of "X" is 0.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
WORD1	PREAMBLE								TLM Message														ISF		R	Parity				
WORD2	TOW(Time Of Week)																	Alert	AS	SubFrameID		PC		Parity						
WORD3	WN(WeekNumber)										C/A OR P OR L2		URA INDEX			SV HEALTH					IODC(L HBNs)		Parity							
WORD4	L2P	Reserved																					Parity							
WORD5	Reserved																					Parity								
WORD6	Reserved																					Parity								
WORD7	Reserved																	TGD(Group delay between SV clock and L1C/A)					Parity							
WORD8	IODC(8 LSBs)								Toc(Epoch of SV clock)														Parity							
WORD9	af2(SV clock drift rate)								af1(SV clock drift)														Parity							
WORD10	af0(SV clock bias)																			PC		Parity								

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
WORD1	PREAMBLE							TLM Message															ISF	R	Parity					
WORD2	RTOW = TOW + X																	Alert	AS	SubFrameID	PC	Parity								
WORD3	WN(WeekNumber)							C/A OR P OR L2			URA INDEX				SV HEALTH					IODC(A MBits)			Parity							
WORD4	L2P	Reserved																		Parity										
WORD5	Reserved																		Parity											
WORD6	Reserved																		Parity											
WORD7	Reserved															TGD(Group delay between SV clock and L1C/A)					Parity									
WORD8	IODC(8 LSBs)							Toc(Epoch of SV clock)															Parity							
WORD9	af2(SV clock drift rate)							af1(SV clock drift)															Parity							
WORD10	af0(SV clock bias)															PC					Parity									

Figure 6-51: Modification of TOW to RTOW from GPS LNAV (Top) to GPS RNAV (Bottom)

6.6.2 RTOW Handling in Receiver

The TOW value in GNSS NAV message received by the receiver will be 5 to 10 minutes ahead of RTOW value in QZSS L6E QZNMA message. This is due to fact that RTOW used the same TOW value while generating DS data for GNSS NAV message at time TOW. However, this DS data is transmitted 5 to 10 minutes later using L6E signal in QZNMA message. Thus, at the receiver, when we receive GNSS NAV message, the TOW values will be 5 to 10 minutes ahead of RTOW value. In order to compute HASH from GNSS NAV message this TOW value must be replaced by RTOW value from the QZNMA message. The pseudo algorithm to do this will be as follows:

Case 1: When memory is not used to store GNSS NAV Message

Step 1: Get RTOW from QZSS L6E QZNMA message

Step 2: Get TOW from GNSS NAV message (LNAV, CNAV, CNAV2 of GPS or I/NAV, F/NAV of Galileo)

Step 3: Replace the TOW value in GNSS NAV message by RTOW value to generate RNAV data.

Step 4: Use the generated RNAV data to compute HASH

Step 5: Use the HASH data for authentication check by using the DS data from QZNMA and public key.

Advantage: Low resource consumption

Disadvantage: Galileo can't be verified

Case 2: When memory is used to store GNSS NAV Message

Step 1: Memory storage is between X to X+5 minutes

Step 2: Search target GNSS NAV message that matches RTOW to TOW

Step 3: Compute the DS using the selected NAV message.

Advantage: Galileo can be verified

Disadvantage: High resource consumption.

6.7 GNSS Digital Signature Generation Frequency

The MCS generates digital signature data for GPS and Galileo signals at every 5 minutes interval for all 32 GPS satellites and 30 Galileo satellites. A total of 156 DS data are generated at the MCS. Total DS data for each satellite system is shown in Table 6-7.

Table 6-7: Number of DS data computed by MCS for GPS and Galileo

Satellite System	Number of Signals	Number of Satellites	Total Number of DS Data
GPS	3 (LNAV, CNAV, CNAV2)	32	96
Galileo	2 (I/NAV, F/NAV)	30	60

6.8 QZSS L6E QZNMA Format

QZSS L6E message format for QZNMA is shown in Figure 6-52. The QZNMA message is 2000 bits long and consists of a 49-bit header, a 1695-bit data section, and a 256-bit Reed-Solomon code. One frame of L6E message takes one second to transmit. See applicable document [2] for details on QZSS L6E signal definitions.

Two 576 bits of RDS data and 543 bits of reserved data are stored in L6E 1695 bits data part of the L6E QZNMA message. RDS data are generated for each satellite and each signal. Thus, a QZNMA frame contains two RDS data that may be for two GPS satellites or two Galileo satellites or one GPS and one Galileo satellite as shown in Figure 6-52, Figure 6-53 and Figure 6-54.

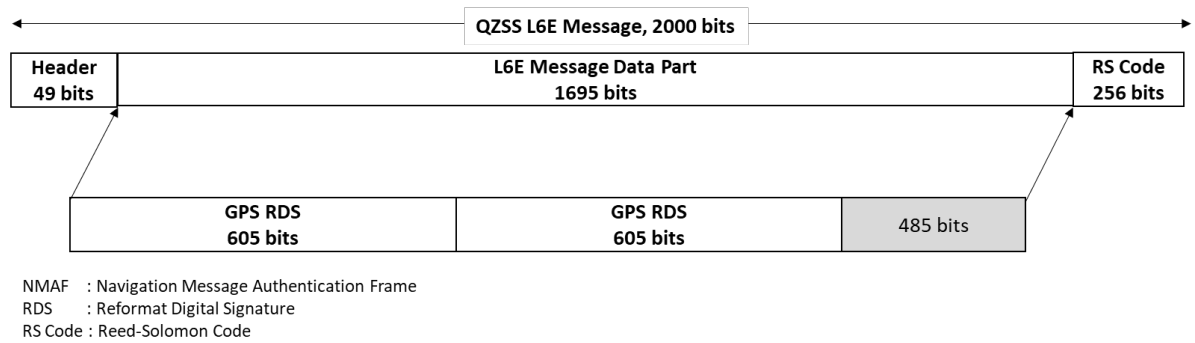


Figure 6-52: QZSS L6E QZNMA Format with Two GPS RDS data

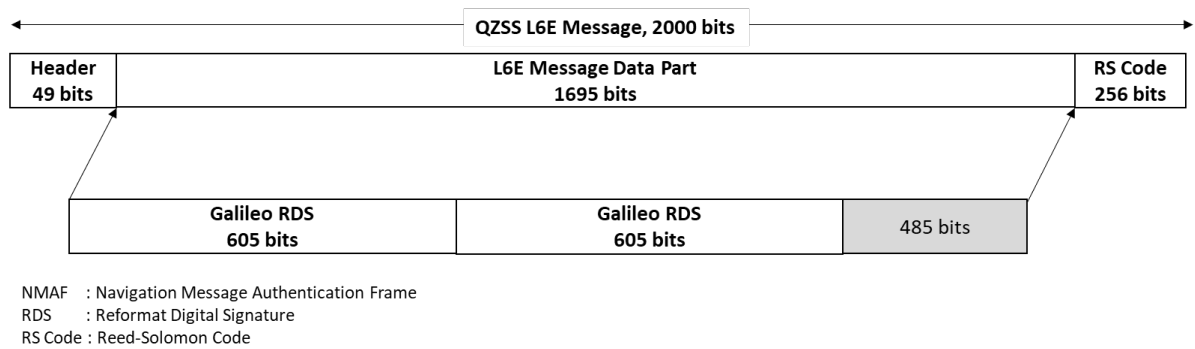


Figure 6-53: QZSS L6E QZNMA Format with Two Galileo RDS data

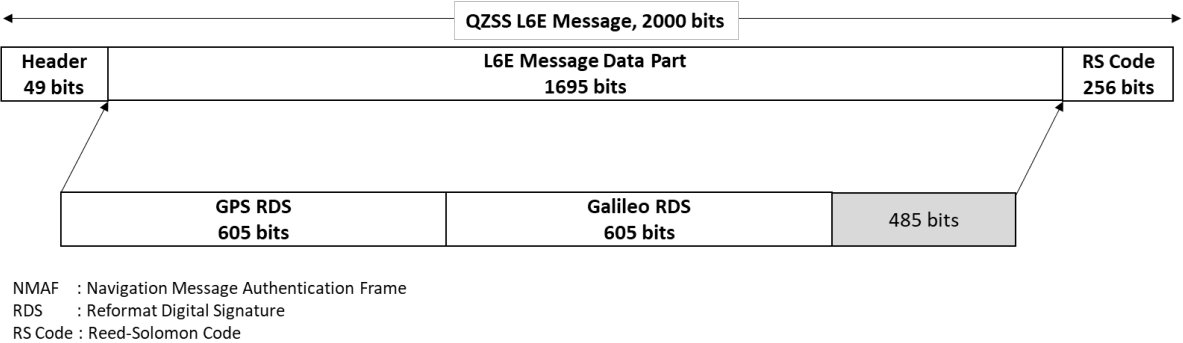


Figure 6-54: QZSS L6E QZNMA Format with One GPS RDS data and One Galileo RDS data

6.9 QZSS L6E QZNMA Broadcast Pattern

QZSS L6E QZNMA broadcast pattern is shown in Figure 6-55. It broadcasts MADOCA-PPP message for high-accuracy services and QZNMA message for signal authentication of GNSS signals. Broadcasting of each message type takes one second. QZNMA message is broadcasted at an interval of every 10 seconds. Thus, two QZNMA messages are broadcasted in 10 seconds. These messages contain four DS data and these four DS data may be for the same satellite with different message types or the same message type of different satellites.

Since DS data are generated at every five minutes interval, all generated DS data are broadcasted within five minutes by using QZSS L6E signal. However, L6E can broadcast only 120 DS data within five minutes duration. Although, a total of 156 DS data are generated, only 120 DS data are broadcasted. The selection of 120 DS data from GPS and Galileo are done by the QZSS L6E message generation system. In case if the MCS computes less than 120 DS data frames, the shortcoming DS data frames of QZNMA are filled with zero values.

See applicable document [2] for details on QZSS L6E message structures.

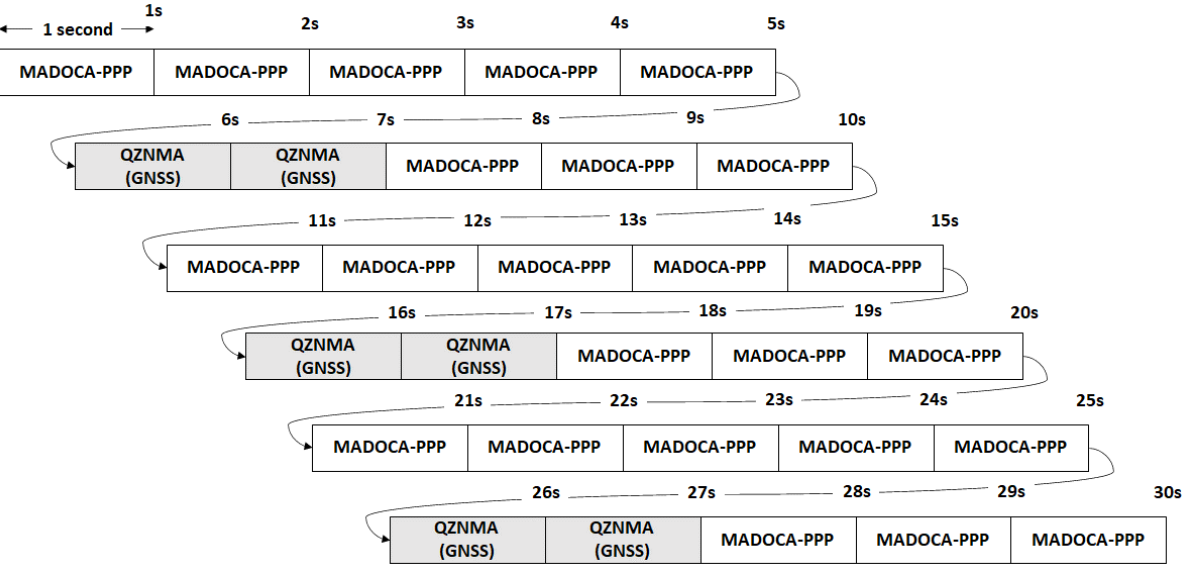


Figure 6-55: QZSS L6E Message Broadcast Pattern

6.10 QZSS L6 Vendor ID and Service Name

Table 6-8 shows Vendor ID and Service Name used in QZSS L6 message to identify message type. QZSS L6 broadcasts two data channels that are named as L6D and L6E. Data channel L6D is used for CLAS service. Data channel L6E is used for MADOCA-PPP and QZNMA services. A receiver identifies L6 message for QZNMA by looking at Vendor ID of the navigation message.

See applicable document [2] for details on QZSS L6 signal.

Table 6-8: Vendor ID and Service Name in L6 Message

Vendor ID	Service Name	Note
101b	CLAS	Broadcast L6D message only
010b	MADOCA-PPP	Broadcast L6D and L6E message
011b	QZNMA	Broadcast L6E message only
Other	Reserved	-

7 Signal Authentication Algorithm

7.1 Digital Signature

7.1.1 Digital Signature Algorithm

ECDSA (Elliptic Curve Digital Signature Algorithm) is used to compute digital signature. See applicable document [8] chapter 6 for details on ECDSA algorithm.

7.1.2 Polynomial Curve

NIST P-256 polynomial curves are used to compute digital signature using ECDSA algorithm. See Appendix D.2.3 of applicable document [8] for polynomial curve details.

Reference: <https://csrc.nist.gov/publications/detail/fips/186/4/final>

NIST P-256 (a.k.a. prime256v1)

P-256 Curve Polynomial: $p = 2^{256} - 2^{224} + 2^{192} + 2^{96} - 1$.

D.1.2.3 Curve P-256	
$p =$	1157920892103562487626974469494075735300861434152903141955 33631308867097853951
$n =$	115792089210356248762697446949407573529996955224135760342 422259061068512044369
$SEED =$	c49d3608 86e70493 6a6678e1 139d26b7 819f7e90
$c =$	7efba166 2985be94 03cb055c 75d4f7e0 ce8d84a9 c5114abc af317768 0104fa0d
$b =$	5ac635d8 aa3a93e7 b3ebbd55 769886bc 651d06b0 cc53b0f6 3bce3c3e 27d2604b
$G_x =$	6b17d1f2 e12c4247 f8bce6e5 63a440f2 77037d81 2deb33a0 f4a13945 d898c296
$G_y =$	4fe342e2 fe1a7f9b 8ee7eb4a 7c0f9e16 2bce3357 6b315ece cbb64068 37bf51f5

Figure 7-1: Test data for ECDSA, Curve P-256

7.2 Hash Method

7.2.1 Hash Function

SHA-256 algorithm is used to compute hash message digest of the input message. See section 6.2 of applicable document [9] for algorithm details.

7.3 Public Key

7.3.1 Number of public Keys

A total of 108 public keys are generated. At a time only one public key is necessary to verify QZSS, GPS and Galileo signals. The same public key is valid for authentication of all signals. The necessary public key is identified by the Key ID (reference4.5) transmitted in each Navigation Authentication Message.

7.3.2 Size of Public Key

Size of Public Key is 512bit.

7.3.3 Public Key File Format

Public key is distributed to users using DER file format. 108 DER files are distributed. The size of each DER file is about 96 Bytes. The total size of 108 keys is about 10.4 Kbytes.

Filename represents Key ID. For example, filename "001.der" represents Key ID 1.

See reference document [10] for details on the DER format.

7.3.4 Public Key Valid Period

In general, the valid period of a public key is one year from the time of its generation. Once a new public key is used, its ID is broadcasted in Key ID (reference4.5). It is recommended that the receiver shall store all valid public keys in the non-volatile memory of the receiver.

7.3.5 Distribution of Public Key

Those who wish to have public keys please confirm the "MICHIBIKI website" of Cabinet Office.

8 Download link

- [1] IS-QZSS-PNT-005, Quasi-Zenith Satellite System Interface Specification Satellite Positioning, Navigation and Timing Service
<https://qzss.go.jp/en/technical/ps-is-qzss/ps-is-qzss.html>
- [2] IS-QZSS-MDC-002, Quasi-Zenith Satellite System Interface Specification Multi-GNSS Advanced Orbit and Clock Augmentation - Precise Point Positioning
https://qzss.go.jp/technical/download/is_qzss_mdc_agree.html
- [3] IS-GPS-200N, NAVSTAR GPS Space Segment/Navigation User Interfaces
<https://www.gps.gov/technical/icwg/IS-GPS-200N.pdf>
- [4] IS-GPS-705J, NAVSTAR GPS Space Segment/ User Segment L5 Interfaces
<https://www.gps.gov/technical/icwg/IS-GPS-705J.pdf>
- [5] IS-GPS-800J , NAVSTAR GPS Space Segment/ User Segment L1C Interfaces
<https://www.gps.gov/technical/icwg/IS-GPS-800J.pdf>
- [6] Galileo OS SIS 2.0, EUROPEAN GNSS (GALILEO) OPEN SERVICE SIGNAL-IN-SPACE INTERFACE CONTROL DOCUMENT
https://www.gsc-europa.eu/sites/default/files/sites/all/files/Galileo_OS_SIS_ICD_v2.0.pdf
- [7] PS-QZSS-004, Quasi-Zenith Satellite System Performance Standard (To be published.)
<https://qzss.go.jp/en/technical/ps-is-qzss/ps-is-qzss.htm>
- [8] FIPS PUB 186-4, FEDERAL INFORMATION PROCESSING STANDARDS PUBLICATION Digital Signature Standard (DSS)
website: <https://csrc.nist.gov/publications/detail/fips/186/4/final>
Document: <https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.186-4.pdf>
- [9] FIPS PUB 180-4, FEDERAL INFORMATION PROCESSING STANDARDS PUBLICATION Secure Hash Standard (SHS)
<https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.180-4.pdf>
- [10] RFC5480, Elliptic Curve Cryptography, Public Key Information
<https://datatracker.ietf.org/doc/html/rfc5480>

9 APPENDIX A: Test Data

Cabinet Office could give the test data only for those who approved to get public keys. See 7.3.5

10 APPENDIX B: Digital Signature Signing and Verification Process

10.1.1 Signing Process

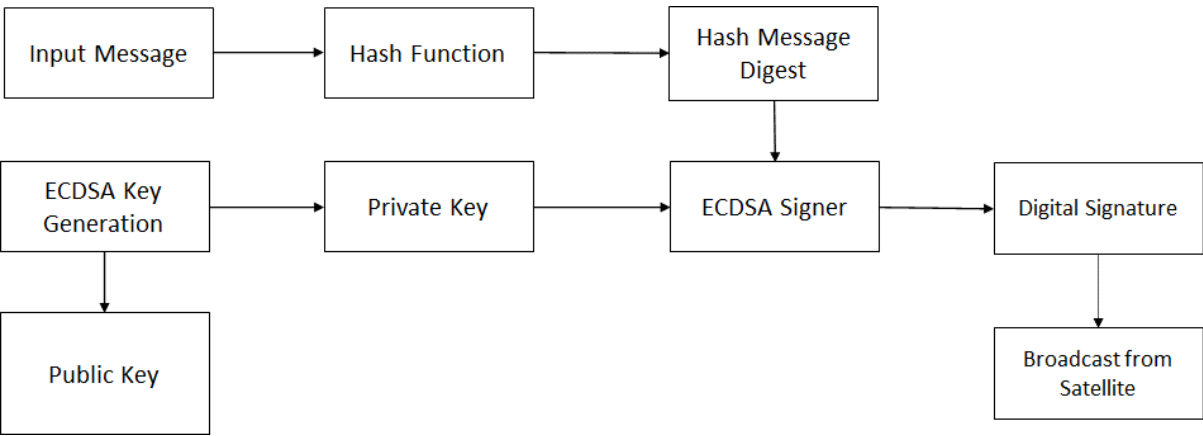


Figure 10-1: Digital Signature signing process

10.1.2 Verification Process

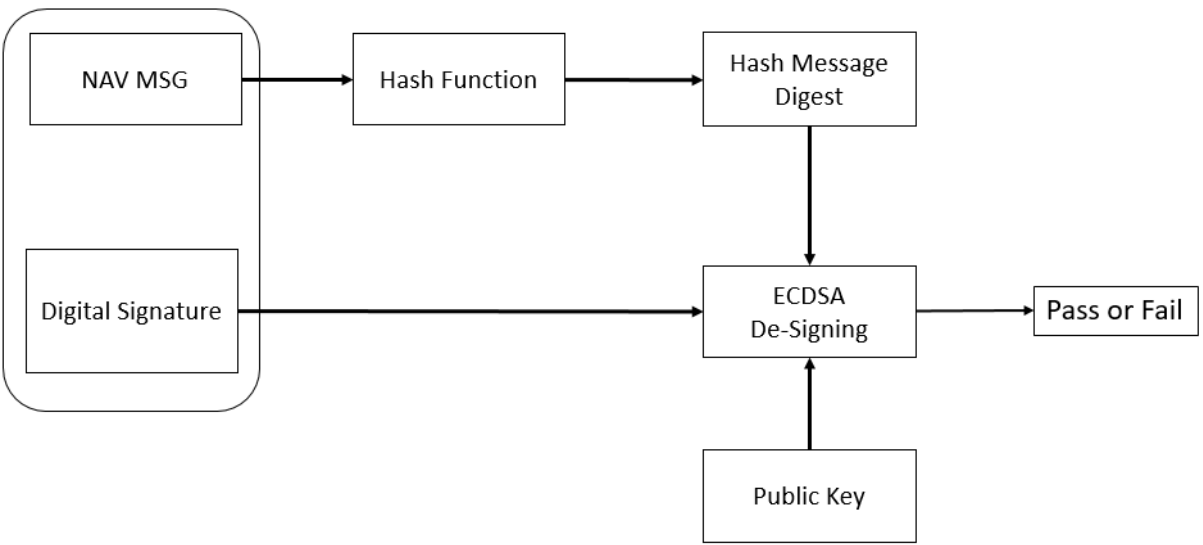


Figure 10-2: Verification process to verify digital signature data